

MODELOS DE CLASIFICACIÓN DE IMÁGENES DE TUMORES DE MAMA

Danillo Barros de Souza



Breast Cancer Project

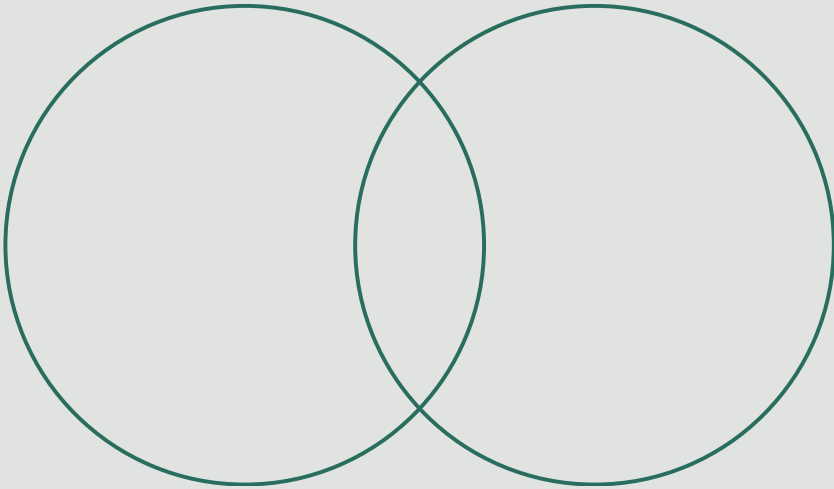


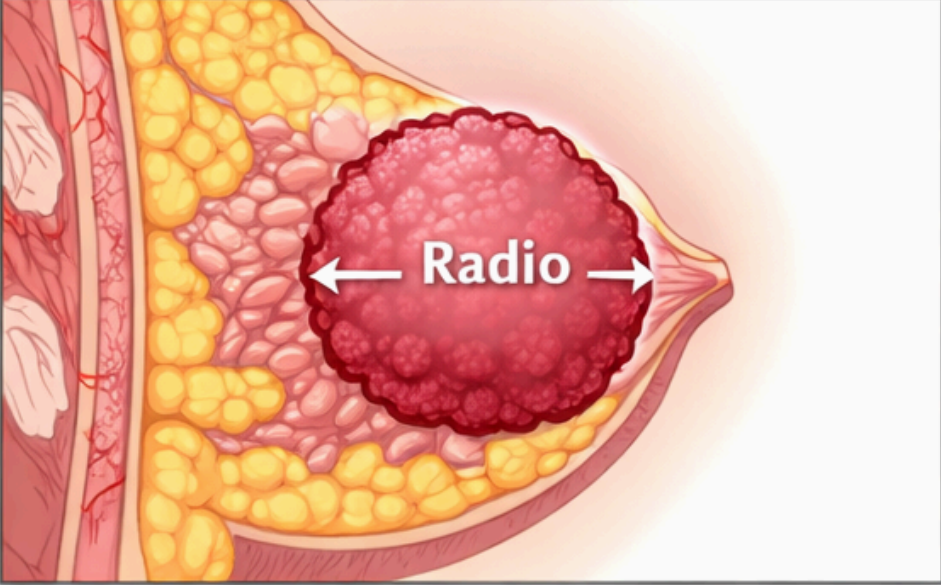
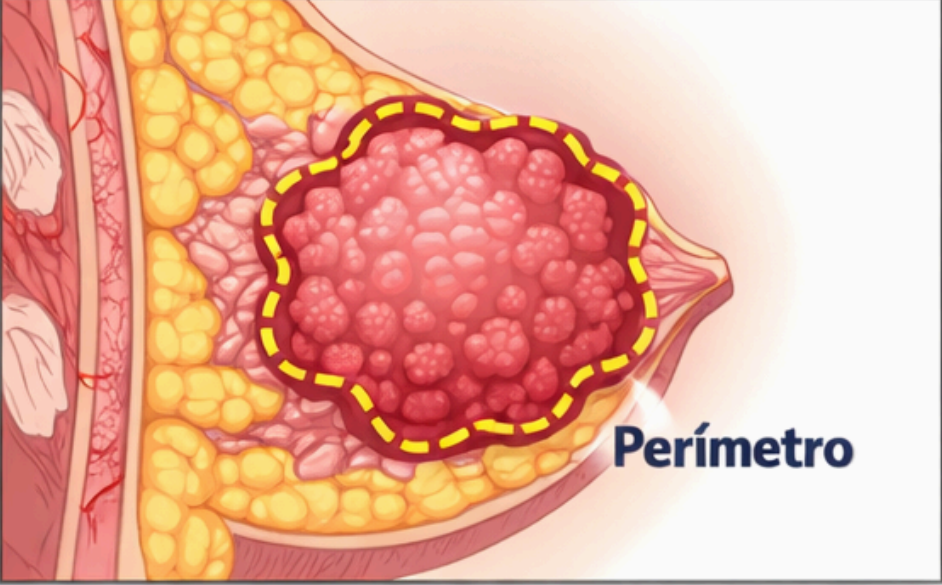
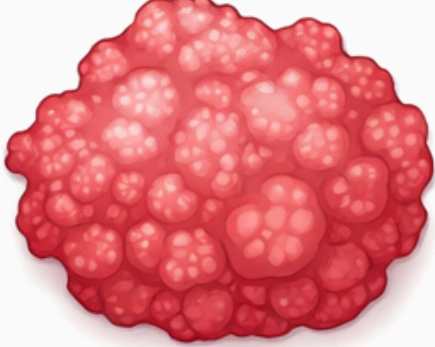
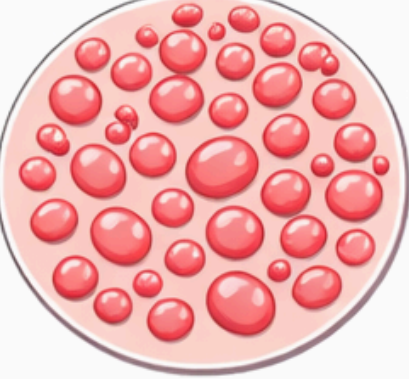
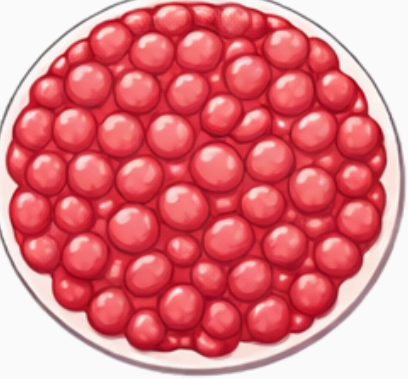
OBJETIVO

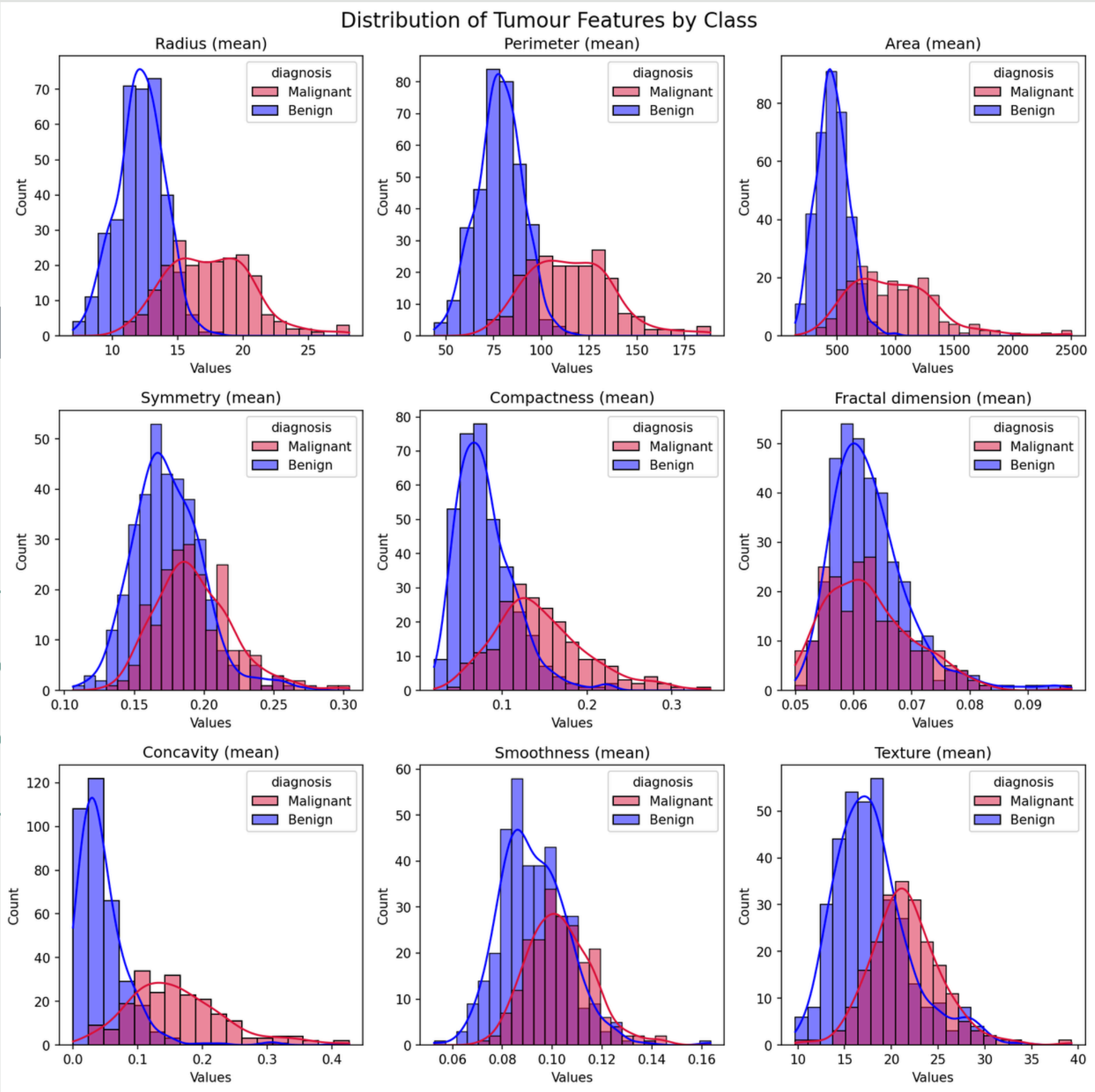
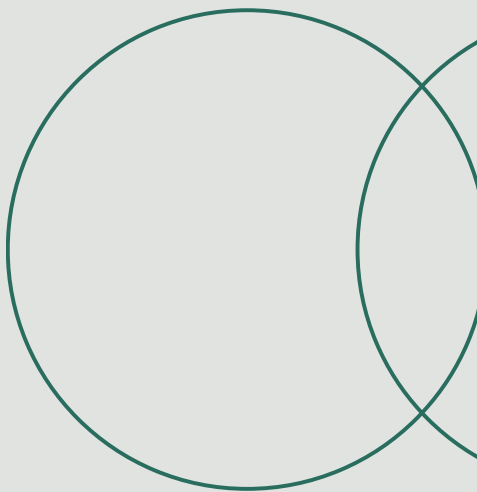


1. Encontrar padrones de tumores a partir de características de imágenes histopatológicas
2. Desplegar un modelo predictivo que utilice informaciones estadísticas de tumores

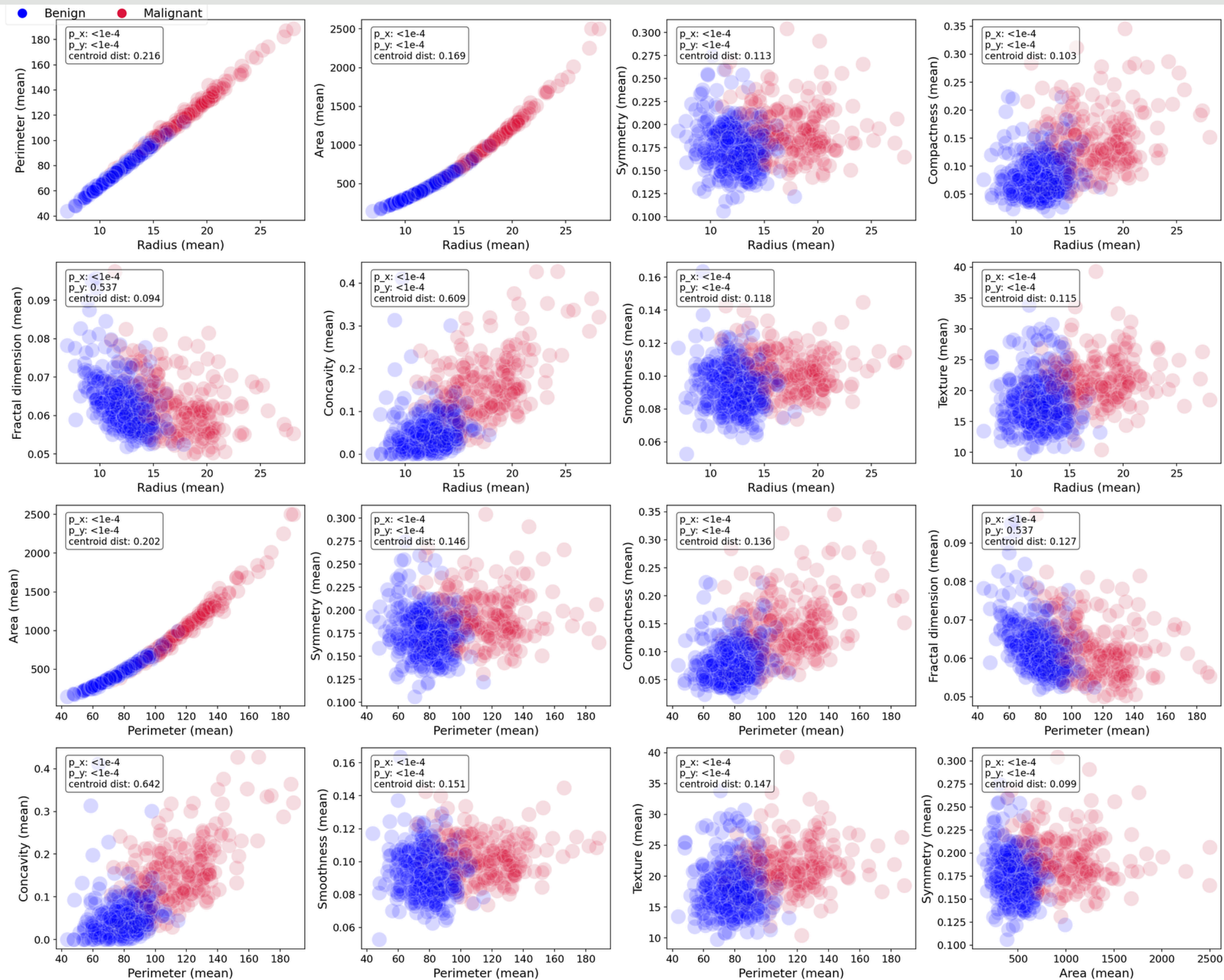
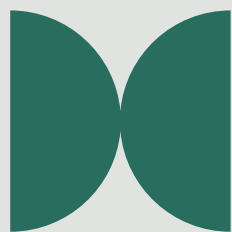
Anteriormente: Características de Contorno



Radio del Tumor Distancia desde el centro al borde	Perímetro del Tumor Longitud del contorito del tumor
	
Textura del Tumor Superficie interna del tumor	Compacidad del Tumor Densidad celular del tumor
Liso  Irregular 	Baja  Alta 

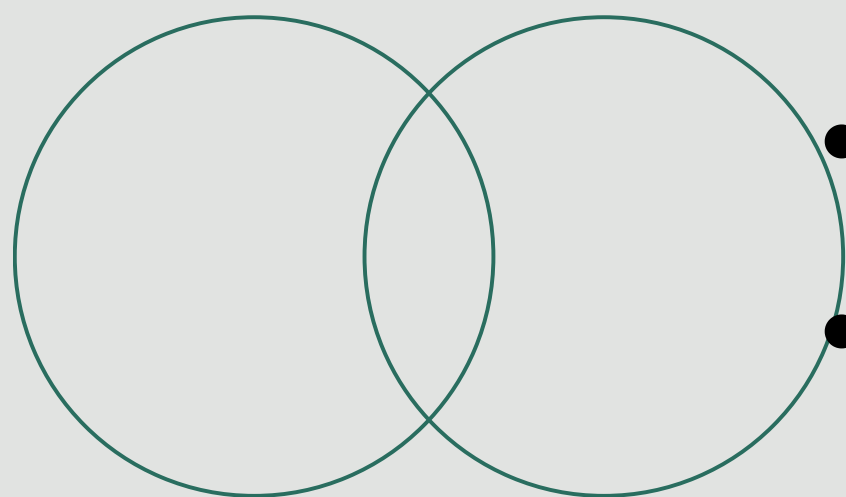


Análisis de Cáncer de Mama

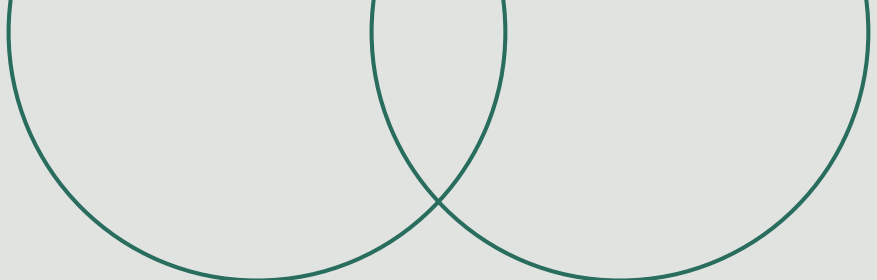




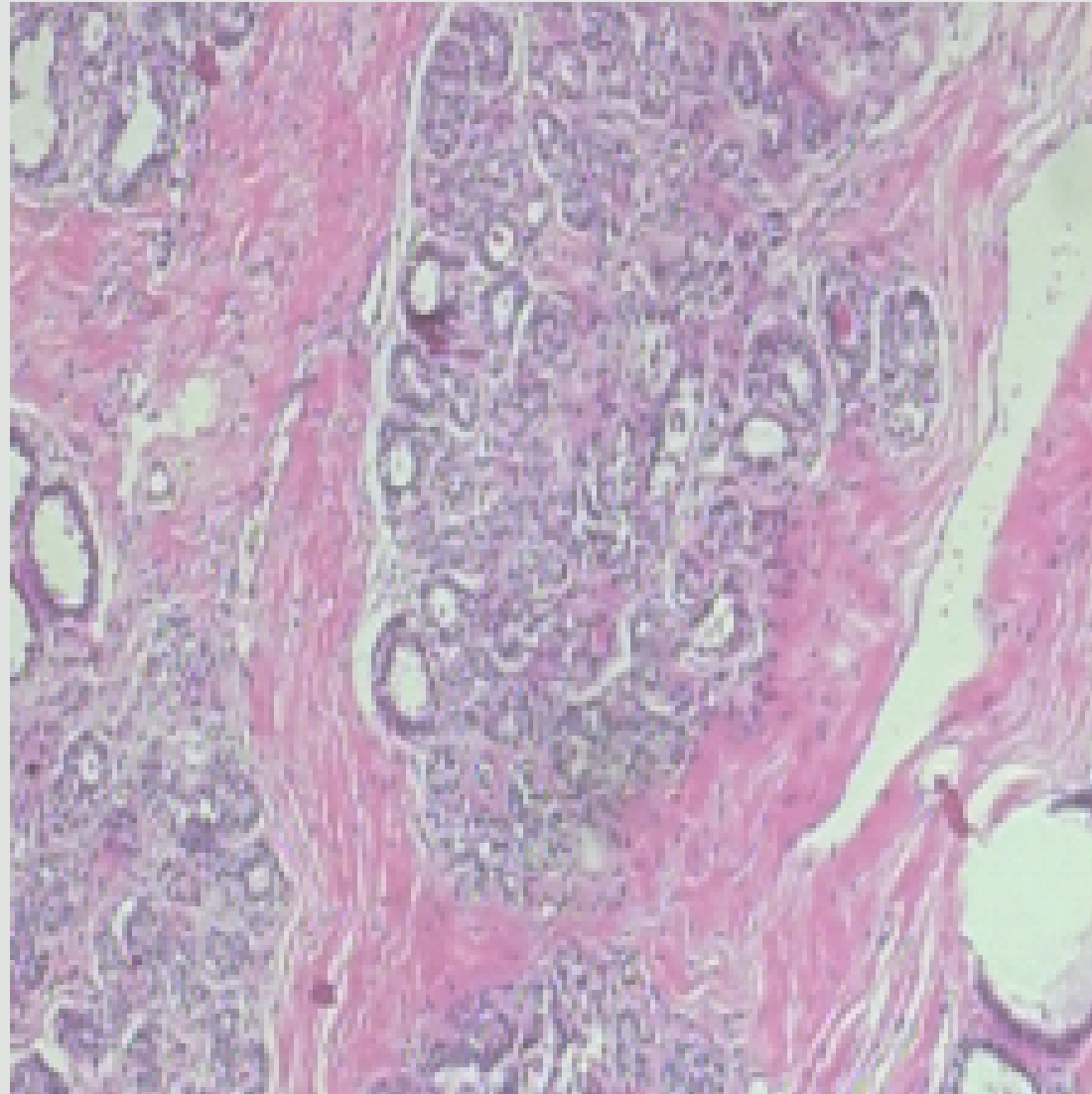
Imágenes de BreakHis



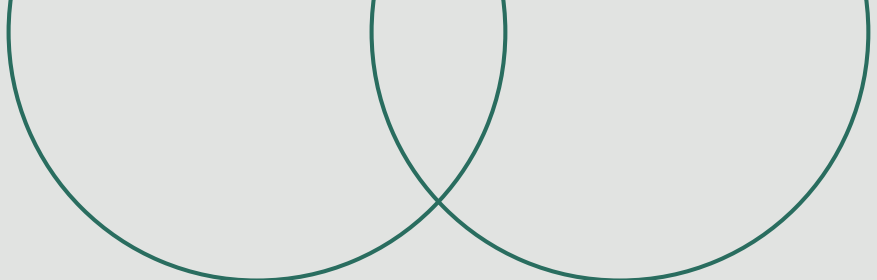
- Dimensiones: 700 x 460
- Magnitud: 40X
- Total: 1995
- Clase 0: 1370
- Clase 1: 625



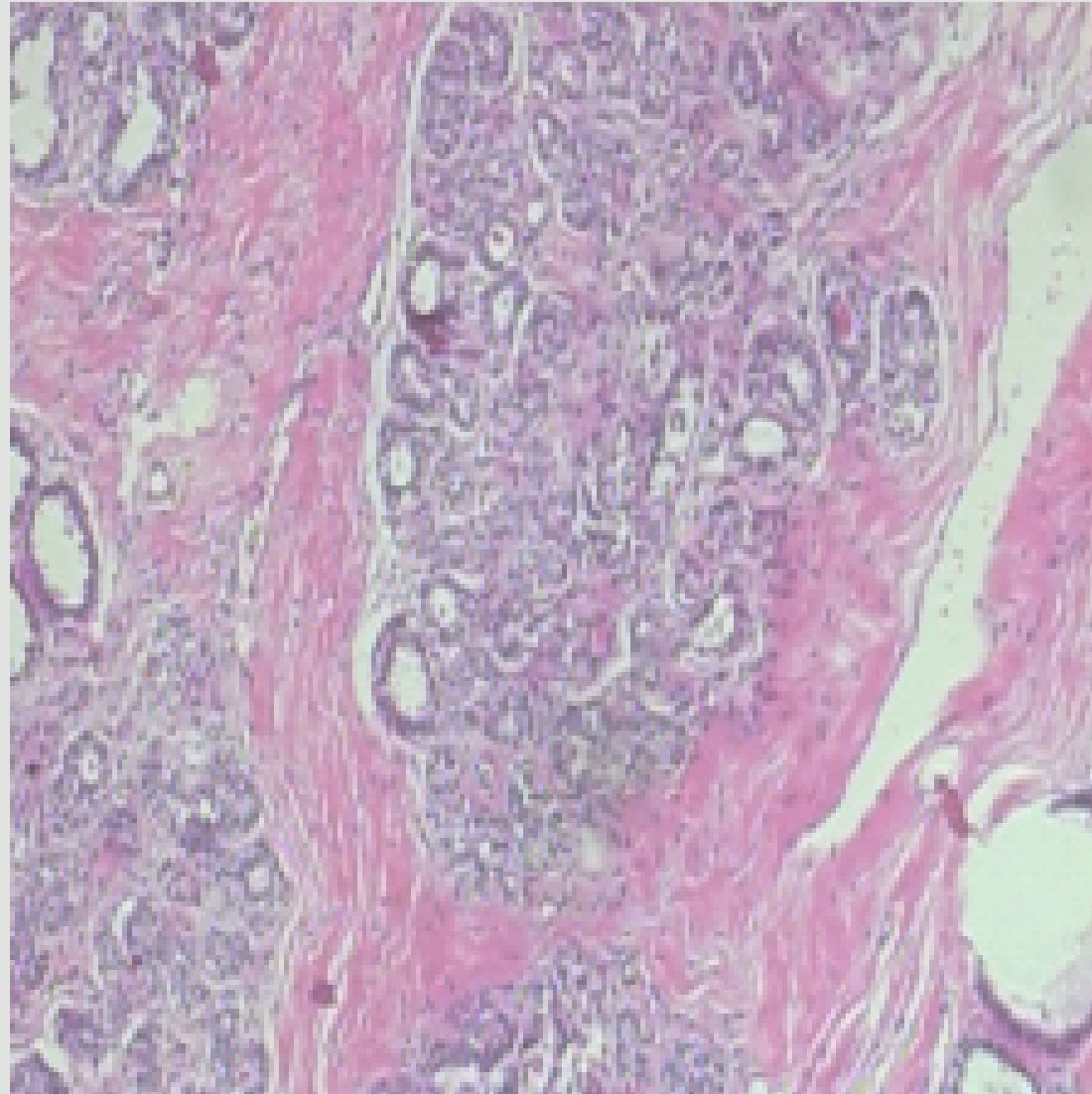
Pregunta: ¿Benigno o Maligno?



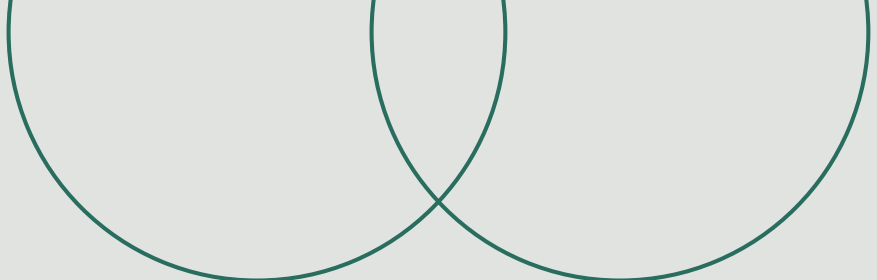
Respuesta:



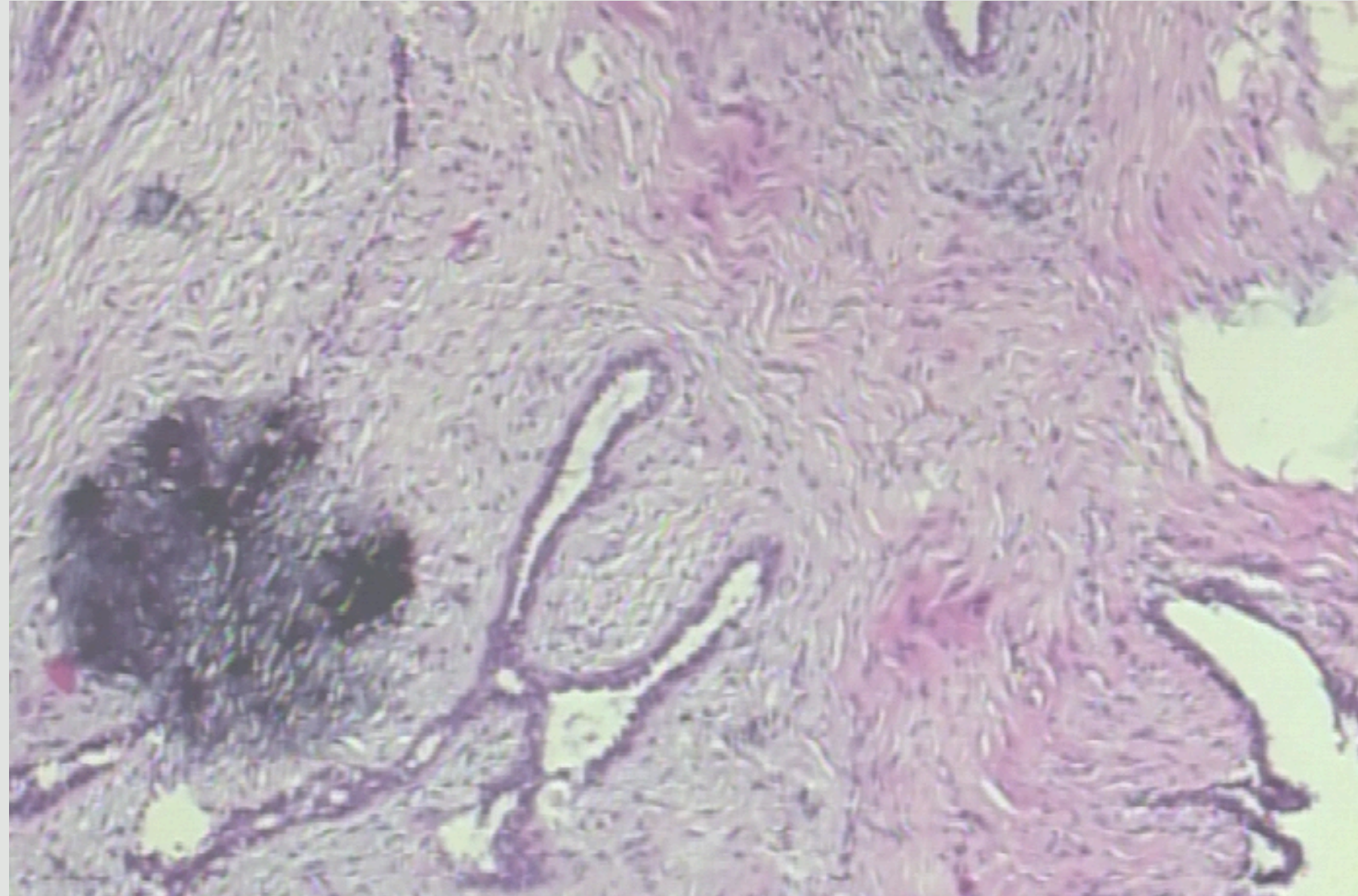
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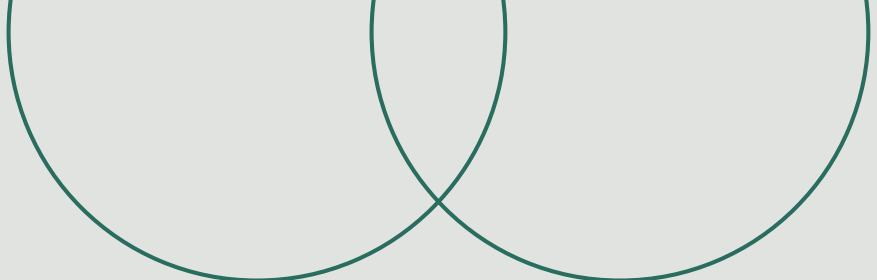


Respuesta: Benigno

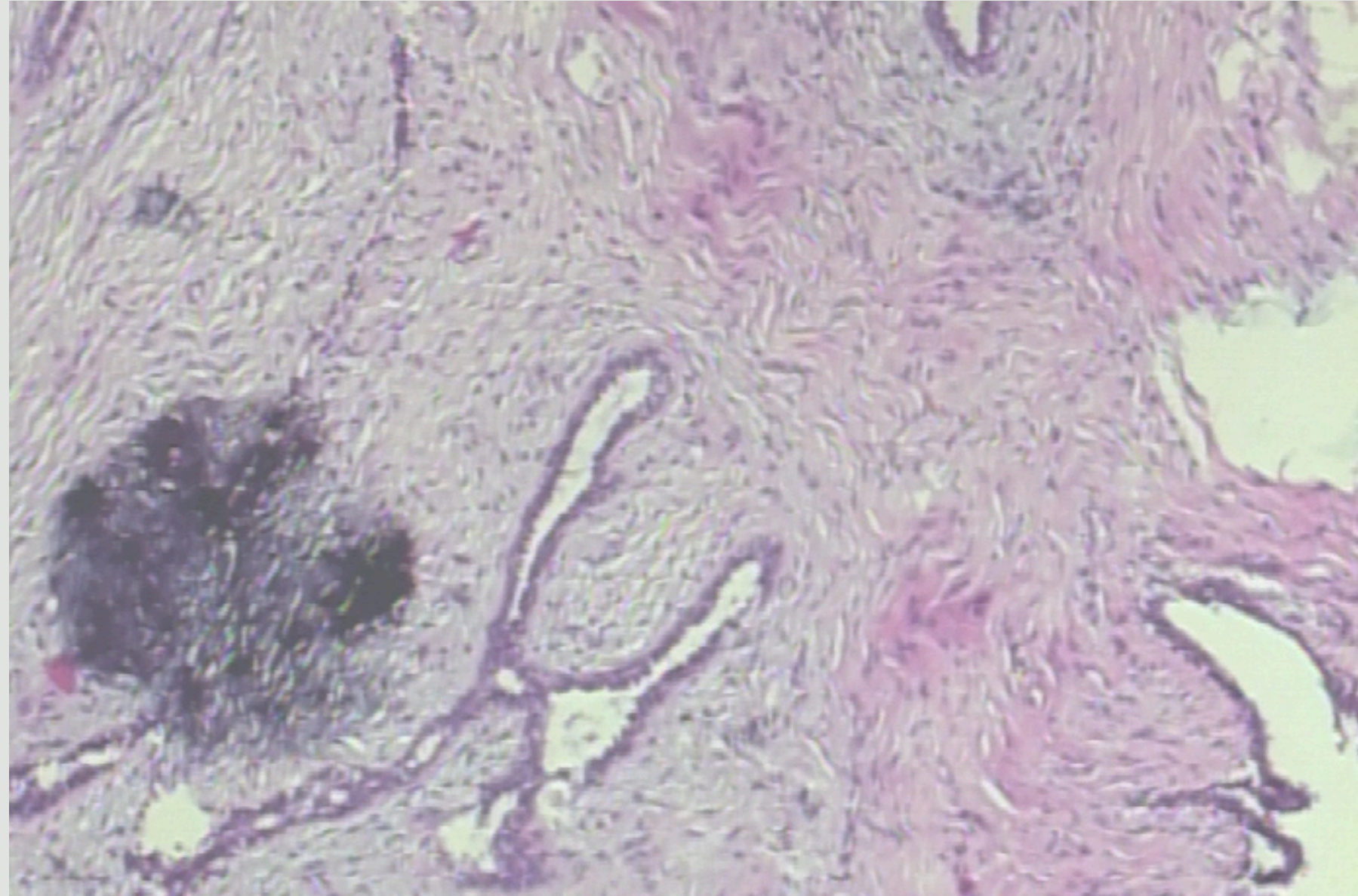


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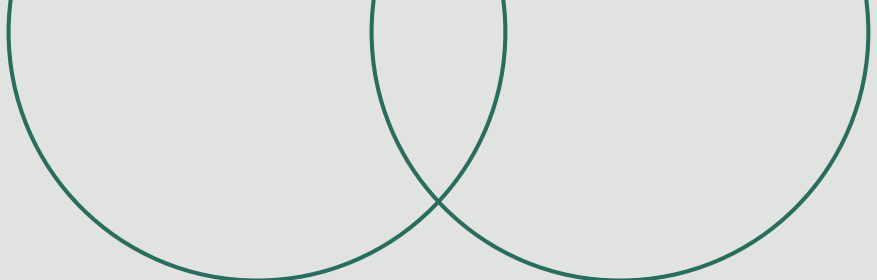




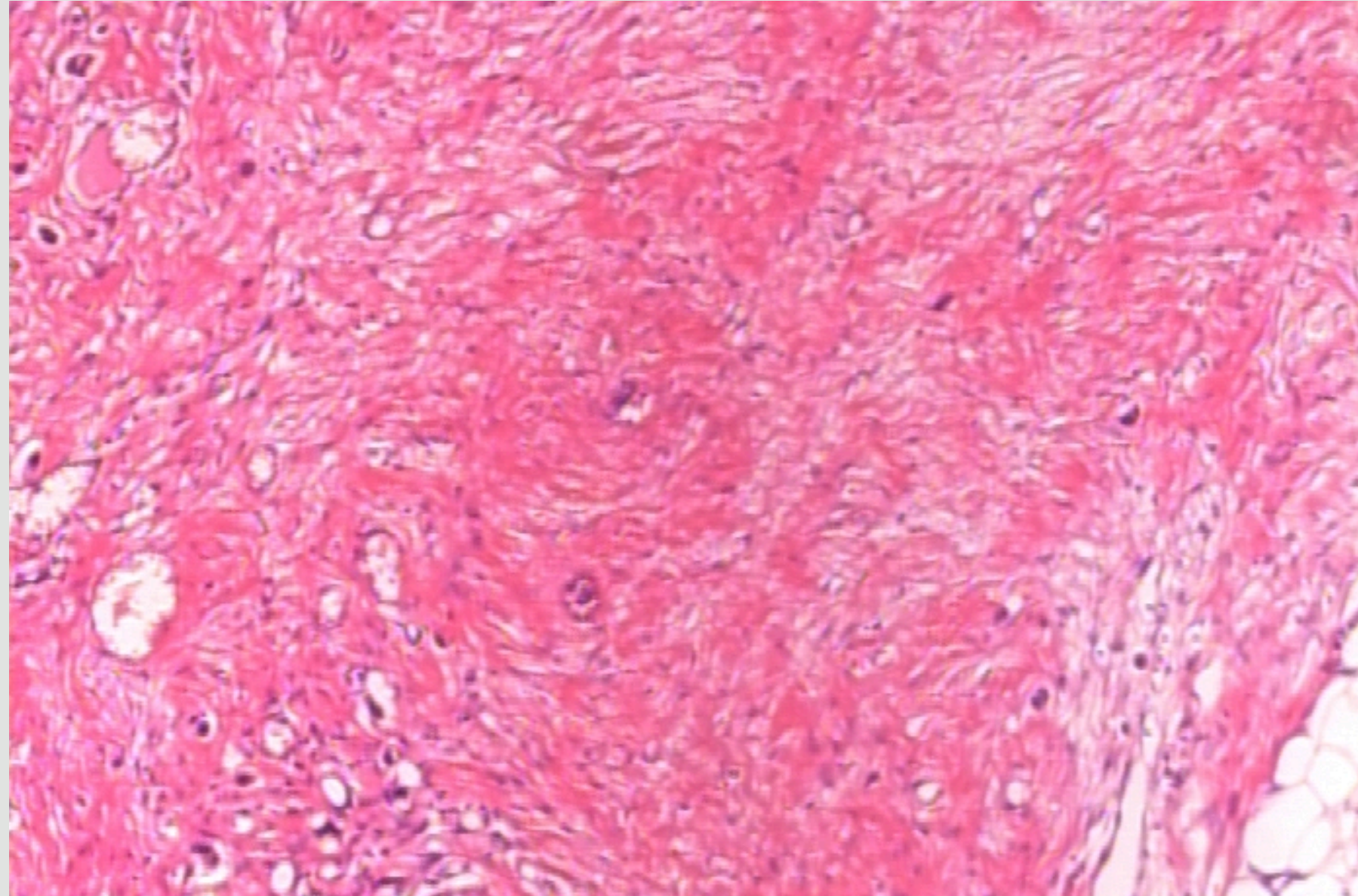
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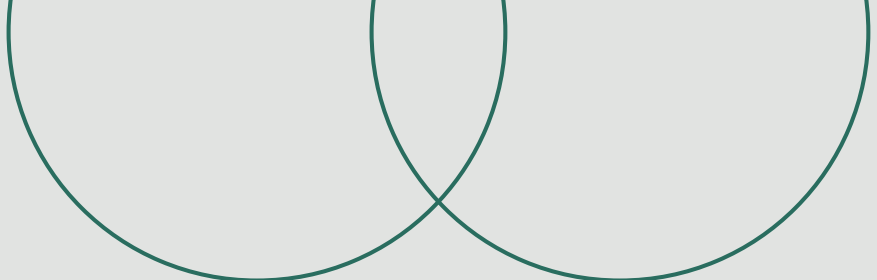


Respuesta: Benigno

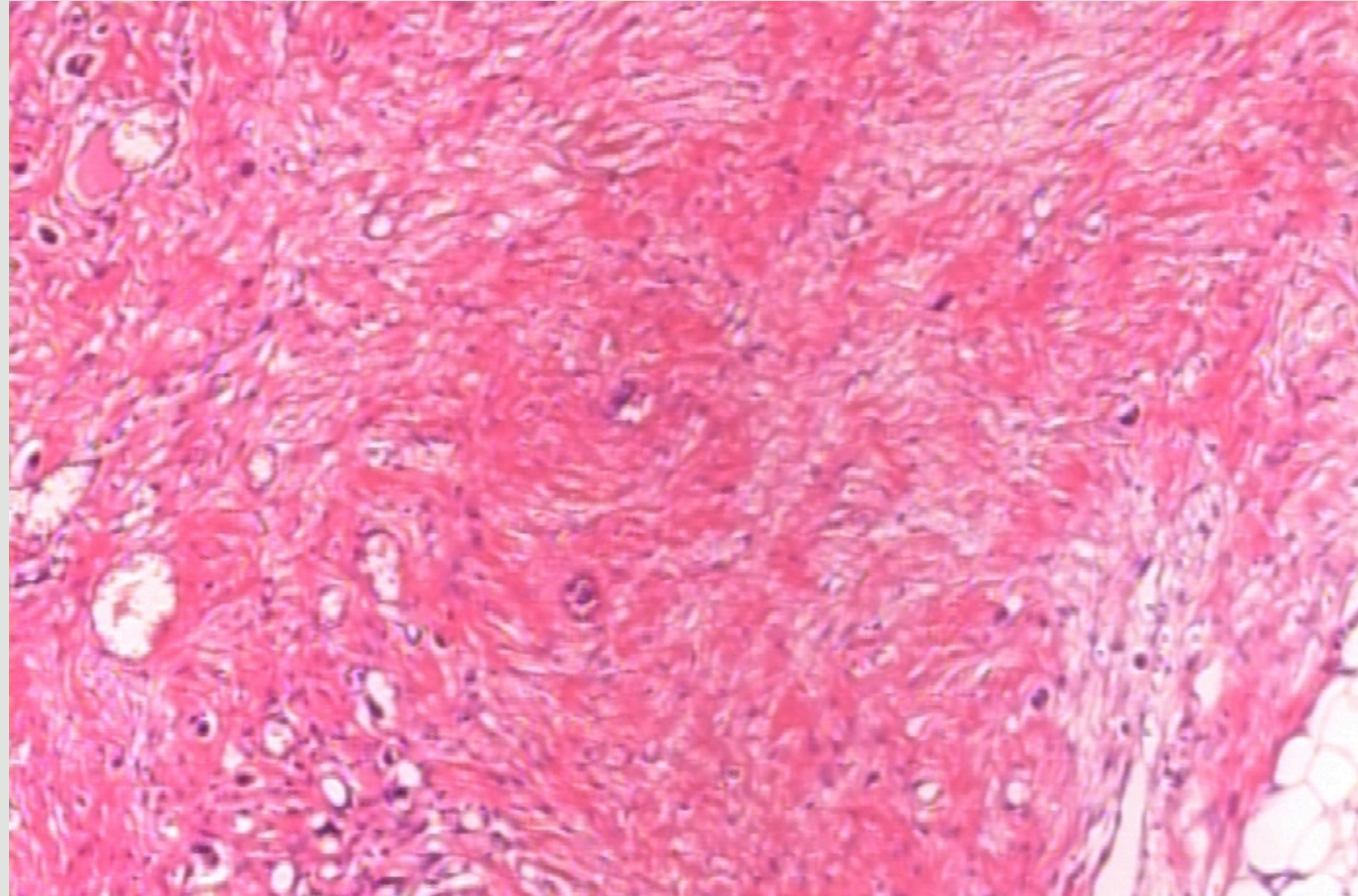


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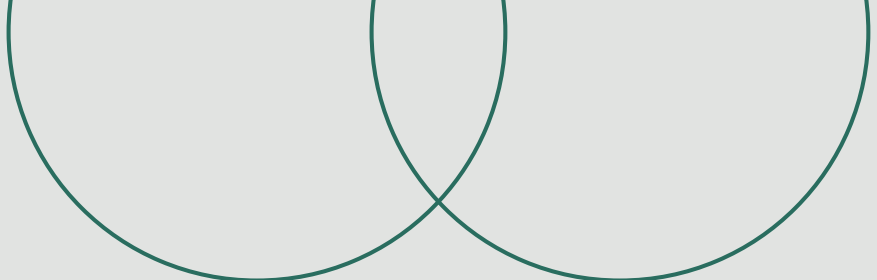




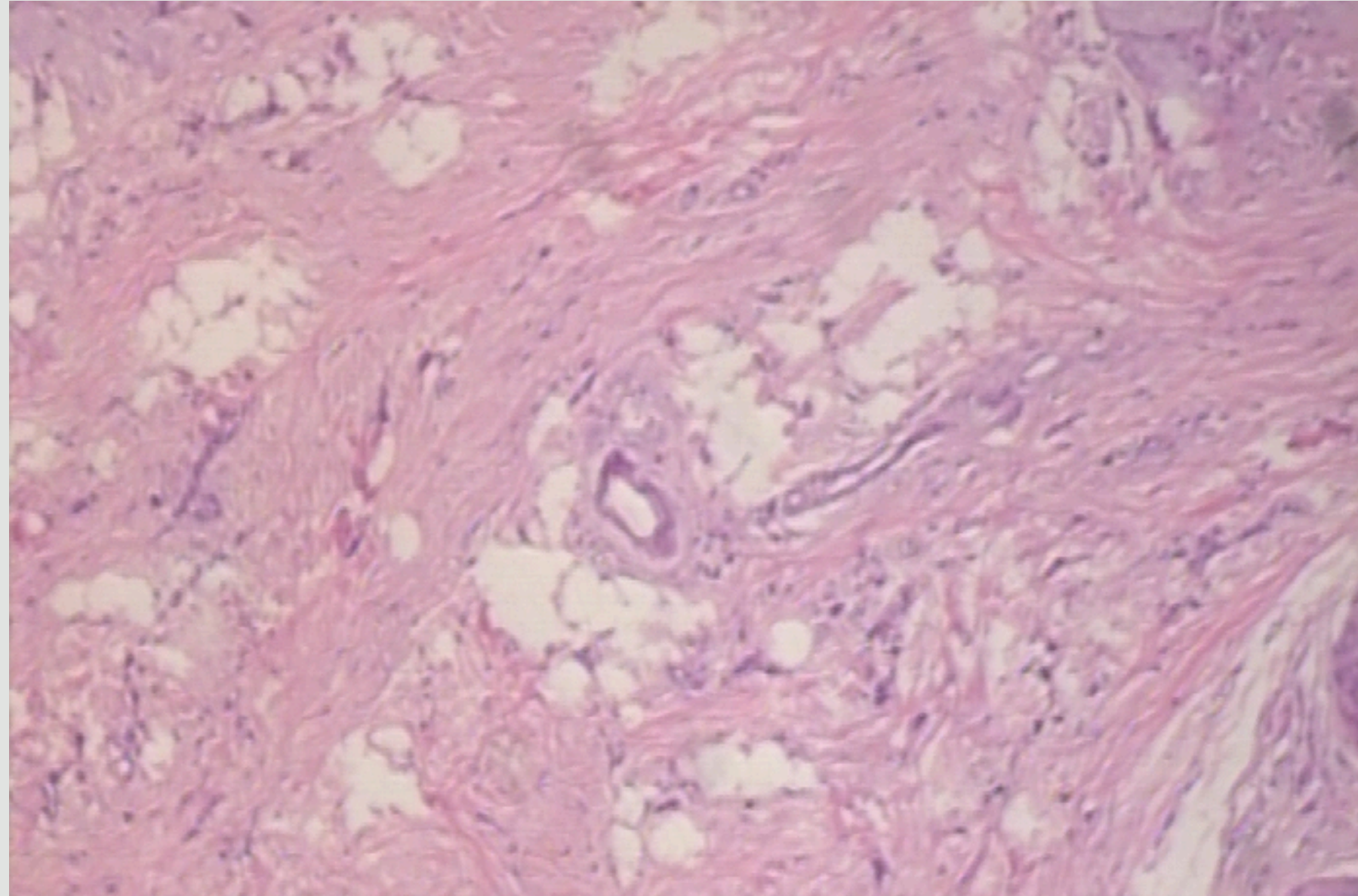
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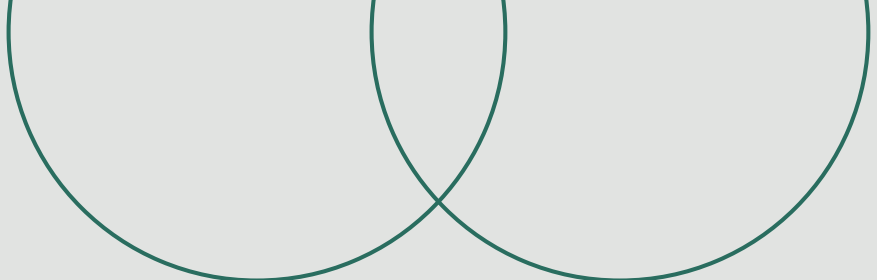


Respuesta: Maligno

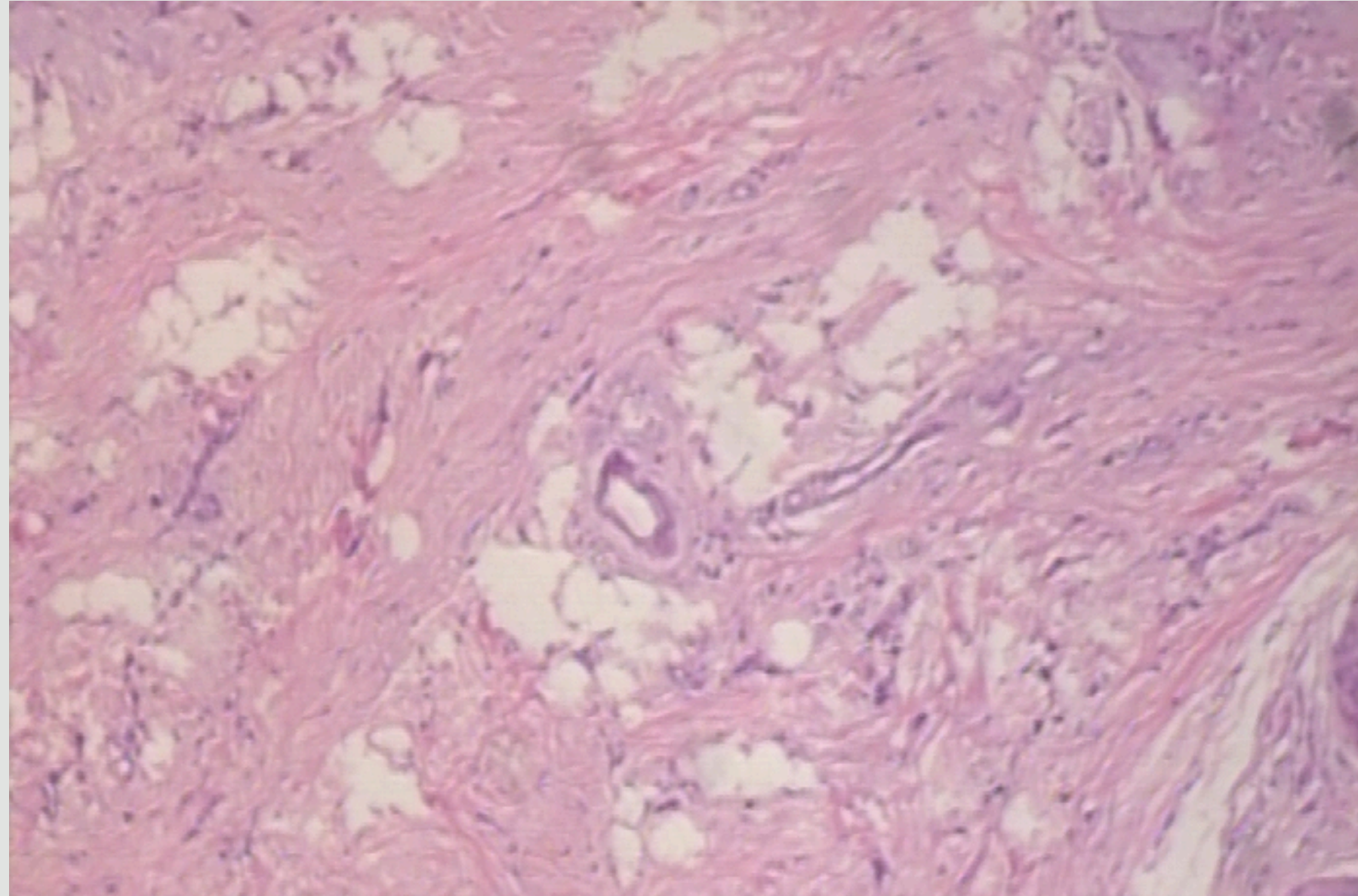


Pregunta: ¿Benigno o Maligno?





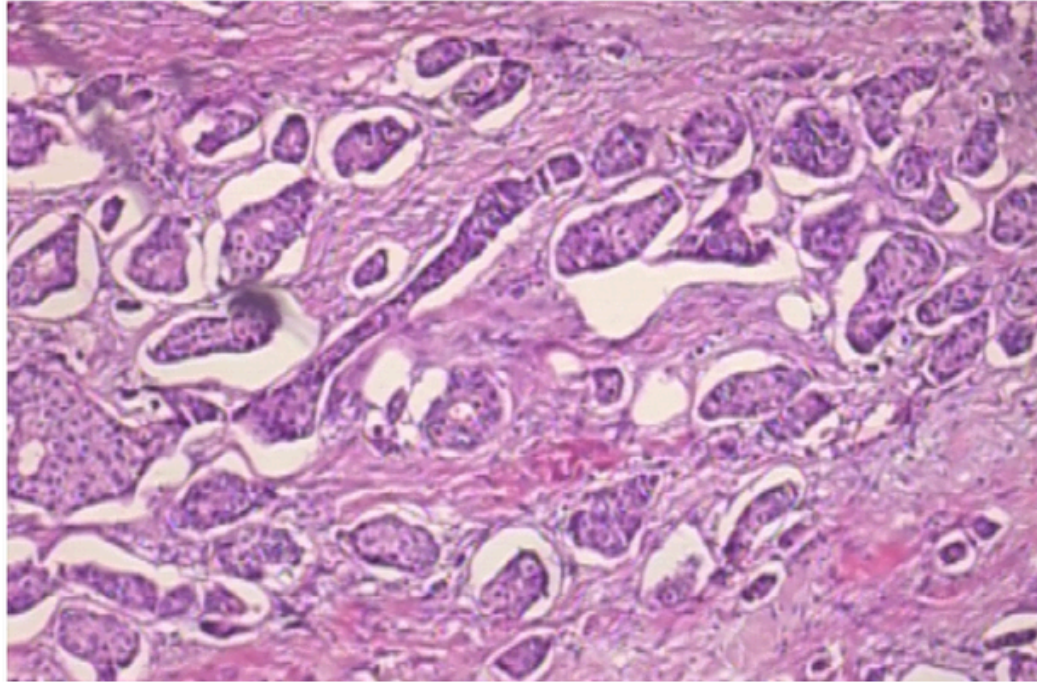
Pregunta: ¿Benigno o Maligno?



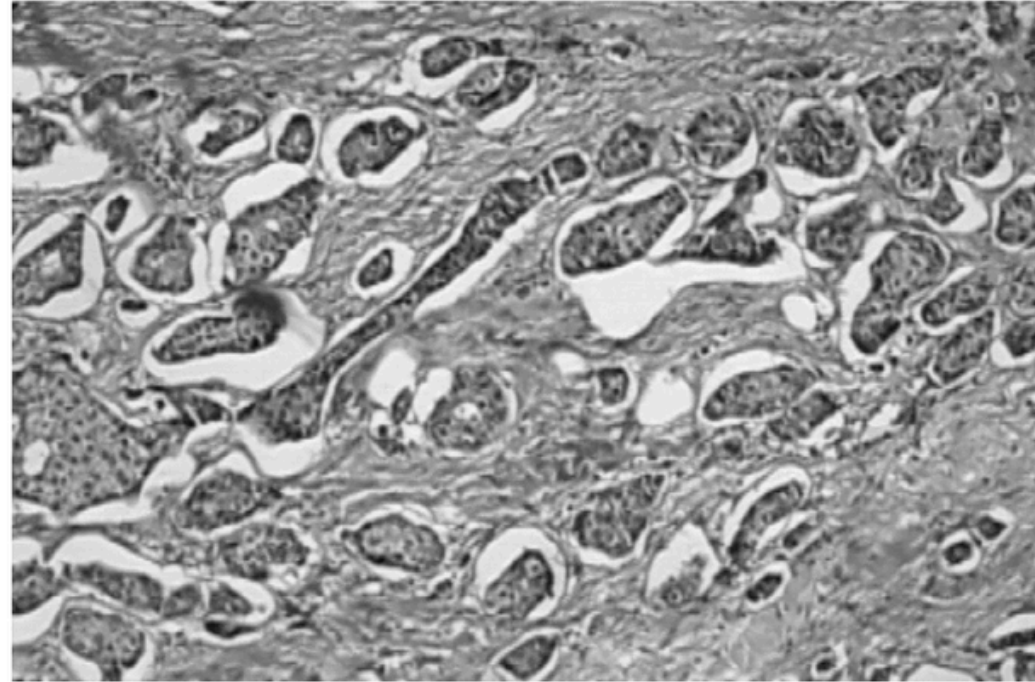
Respuesta: **Maligno**

¿Qué información puedo obtener de una imagen histopatológica?

Original Image



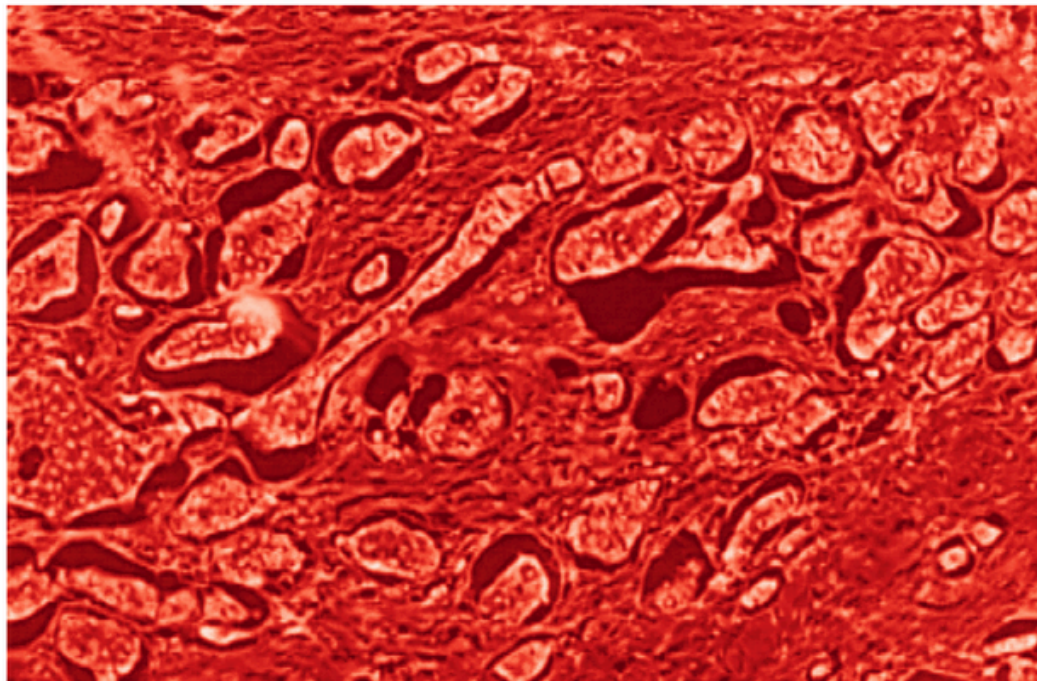
Grayscale Image



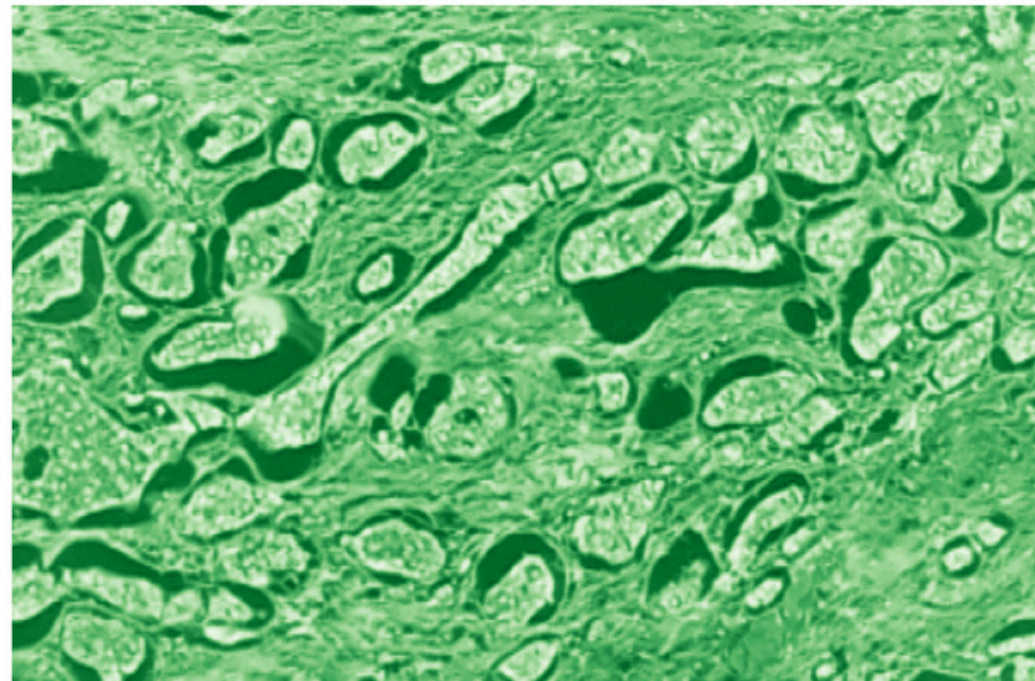
Binary Mask



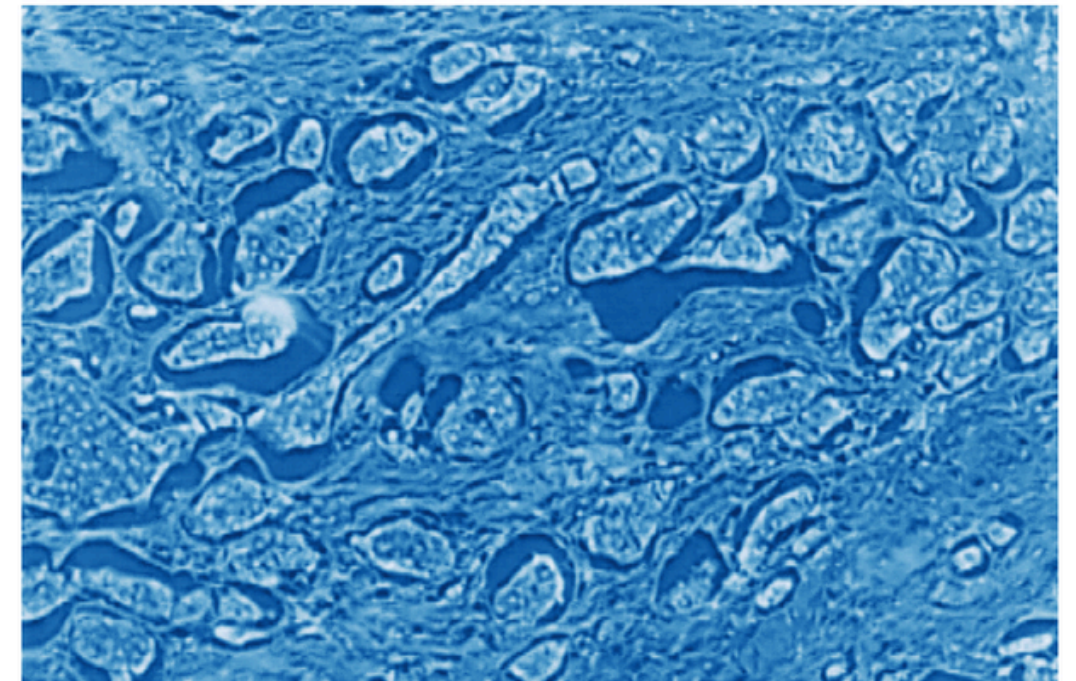
Red Channel



Green Channel



Blue Channel



¿Qué información tengo que utilizar para tener una buena predicción?

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Respuesta:

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**Respuesta:
¡Búscate la vida!**





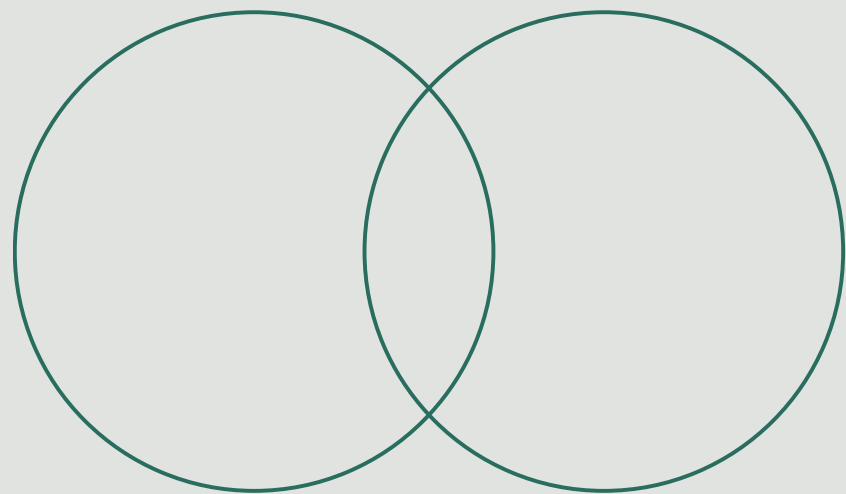
Información Estadística:

Media, Desviación estándar, gradiente, Normal L1,
Transformada de Fourier...



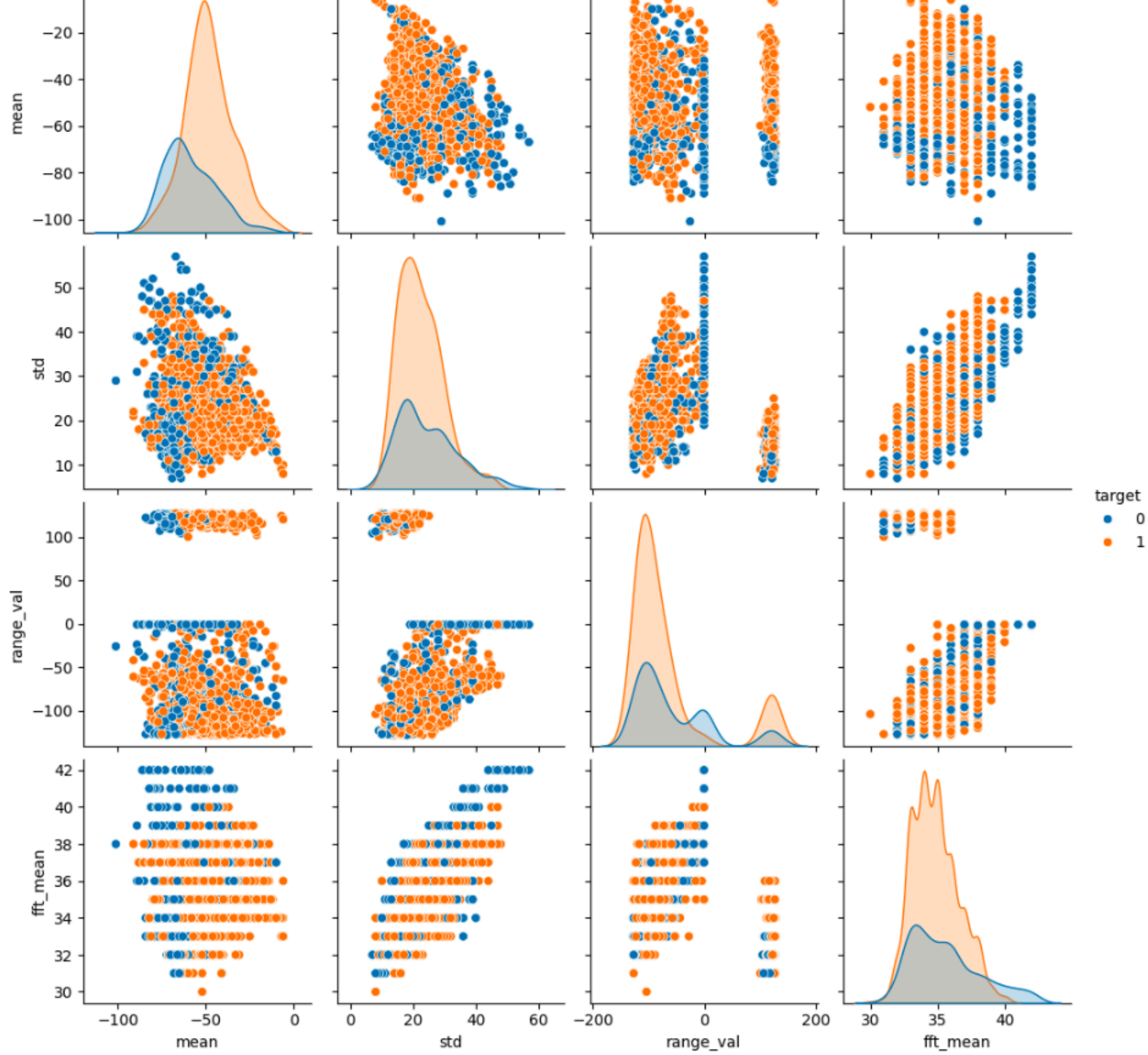
Información visual:

Padrones de color, contraste, textura, contorno...



RESULTADOS EDA (DATOS ESTADÍSTICOS)

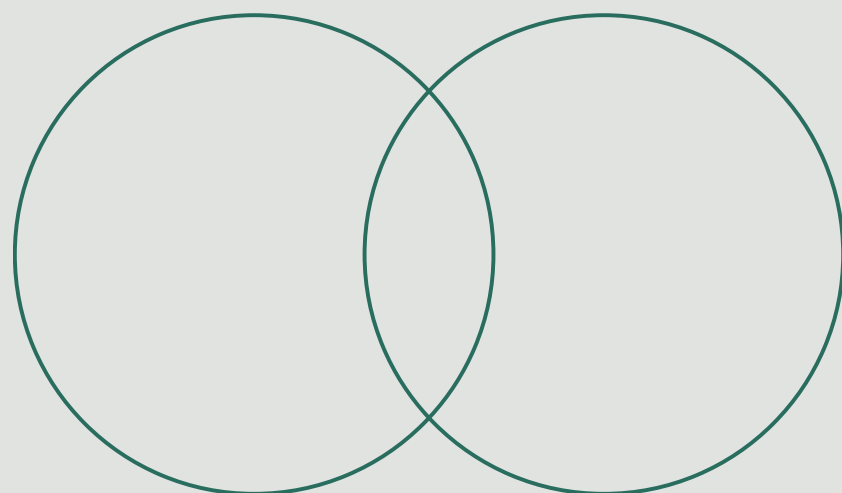
Mejores
resultados:





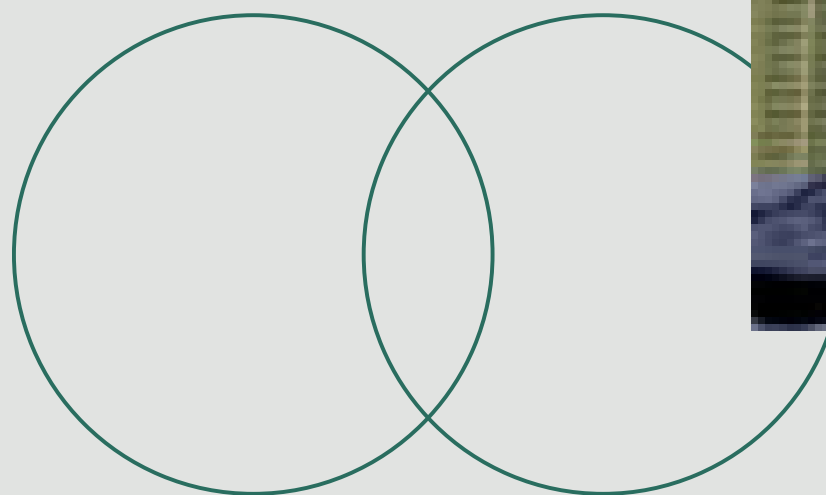


No hay cómo huir de modelos de deep learning :(





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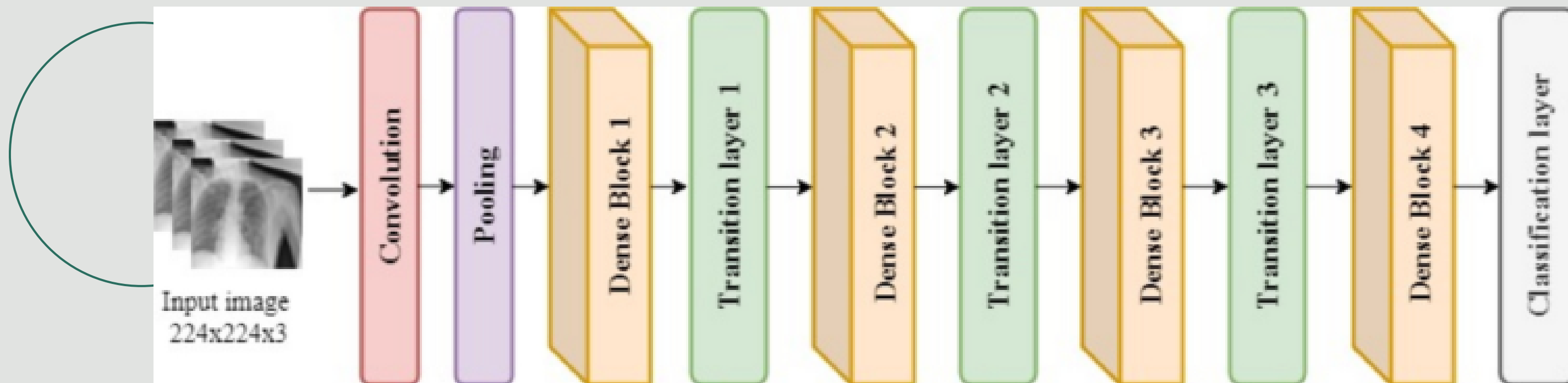


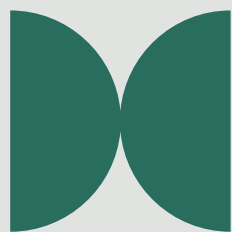
Prompt: “¿Qué modelos preentrenados de deep learning puedo utilizar como base para crear mi modelo de clasificación de histopatología?”



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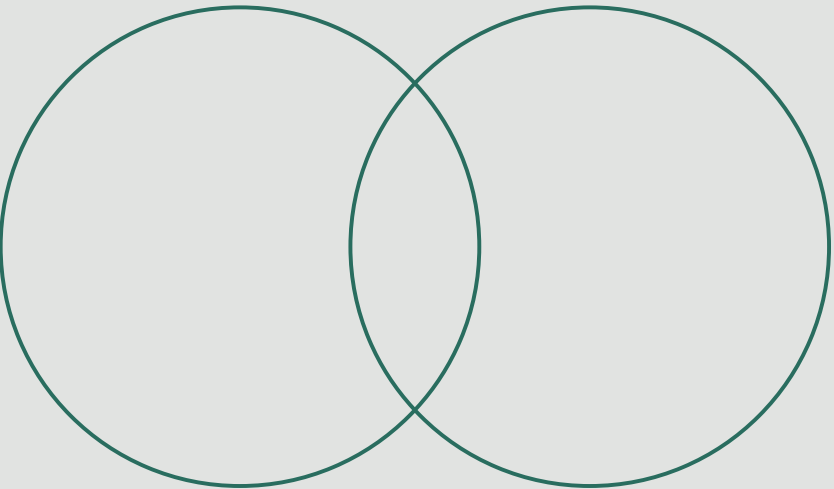
Answer: DenseNet121 (transfer learning)





Modelo Base:

Componente	Configuración
Backbone	DenseNet121
Pretraining Dataset	ImageNet
Include Top Layers	No
Input Resolution	224 × 224
Transfer Learning Strategy	Frozen backbone during initial training



Parámetros ~ 7.000.000

Modelo 0

Capa	Descripción
Input Layer	224 × 224 image
Data Augmentation	Random transformations
DenseNet Preprocessing	ImageNet normalization
DenseNet121 Backbone	Base model
GlobalAveragePooling2D	Spatial feature aggregation
Batch Normalization	Regularization
Dropout	Overfitting reduction
Dense Classification Layer	Feature compression (256)
Sigmoid Output	Binary classification

Modelo 0

Capa	Descripción
Input Layer	224 × 224 image
Data Augmentation	Random transformations
DenseNet Preprocessing	ImageNet normalization
DenseNet121 Backbone	Base model
GlobalAveragePooling2D	Spatial feature aggregation
Batch Normalization	Regularization
Dropout	Overfitting reduction
Dense Classification Layer	Feature compression (256)
Sigmoid Output	Binary classification

Modelo 2

Capa	Descripción
Input Layer	224 × 224 image
Data Augmentation	Random transformations
DenseNet Preprocessing	ImageNet normalization
DenseNet121 Backbone	Base model
GlobalAveragePooling2D	Spatial feature aggregation
Batch Normalization	Regularization
Dropout	Overfitting reduction
Dense Classification Layer	Feature compression (64)
Sigmoid Output	Binary classification

Modelo 1

Capa	Descripción
Input Layer	224 × 224 image
Data Augmentation	Random transformations
DenseNet Preprocessing	ImageNet normalization
DenseNet121 Backbone	Base model
GlobalAveragePooling2D	Spatial feature aggregation
Batch Normalization	Regularization
Dropout	Overfitting reduction
Dense Classification Layer	Feature compression (128)
Sigmoid Output	Binary classification

Modelo 3

Capa	Descripción
Input Layer	224 × 224 image
Data Augmentation	Random transformations
DenseNet Preprocessing	ImageNet normalization
DenseNet121 Backbone	Base model
GlobalAveragePooling2D	Spatial feature aggregation
Batch Normalization	Regularization
Dropout	Overfitting reduction
Dense Classification Layer	Feature compression (32)
Sigmoid Output	Binary classification

Resumen

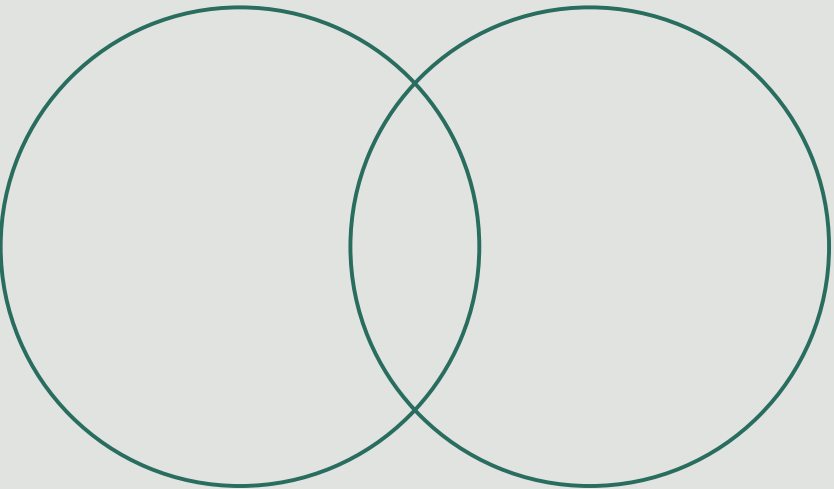
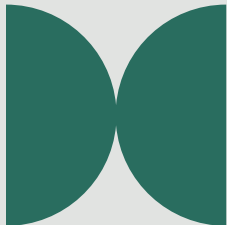
Model	Dense Layer Size	Dropout Rate	Objective
Model 0 (Original)	256 neurons	0.3	Baseline architecture
Model 1 (1st Refinement)	128 neurons	0.15	Reduced complexity
Model 2 (2nd Refinement)	64 neurons	0.15	Further parameter reduction
Model 3 (3rd Refinement)	32 neurons	0.15	Final optimized configuration

Parámetros de entrenamiento

Parameter	Value
Optimizer	Adam
Learning Rate	1×10^{-3}
Loss Function	Binary Crossentropy
Epochs	15
Batch Size	32

Métricas de optimización

Metric	Purpose
Recall	Sensitivity to malignant cases
AUC (ROC-AUC)	Classification separability

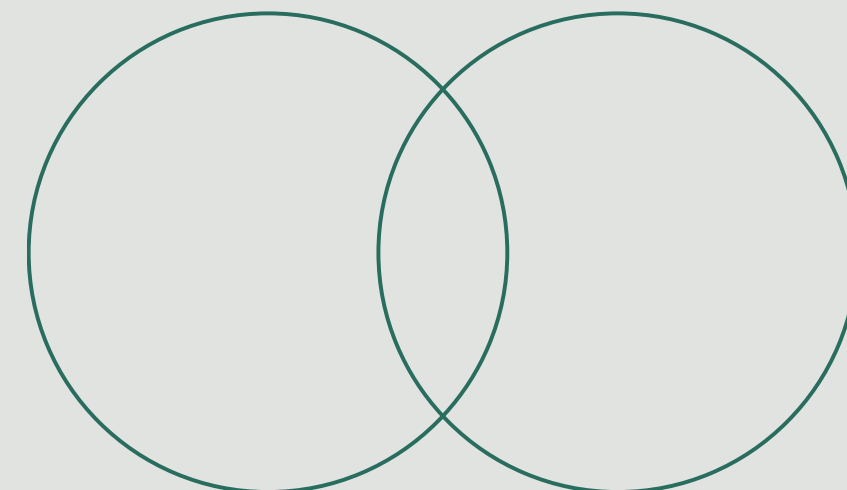




Let the game begin.

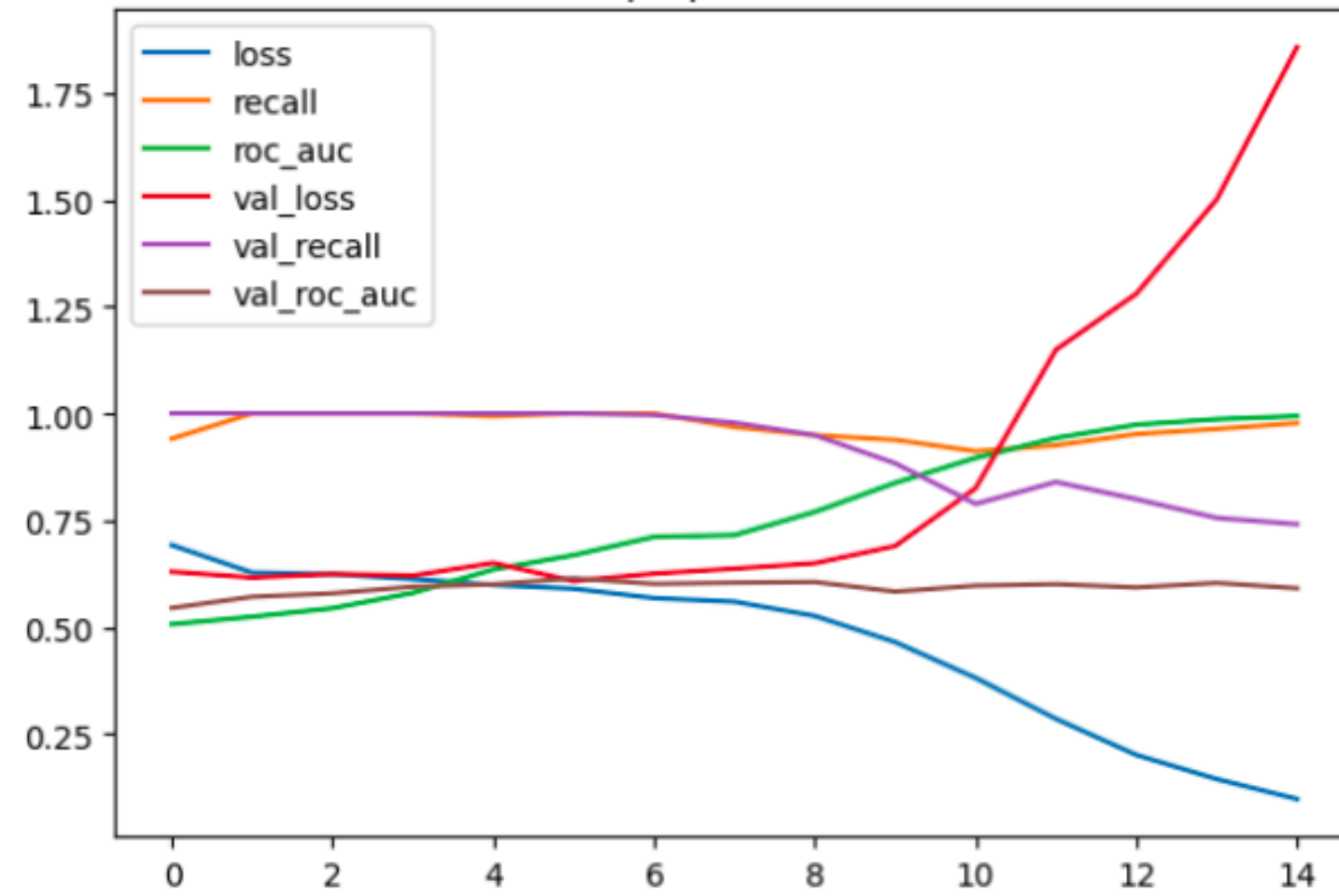


Inputs: Contorno de imágenes

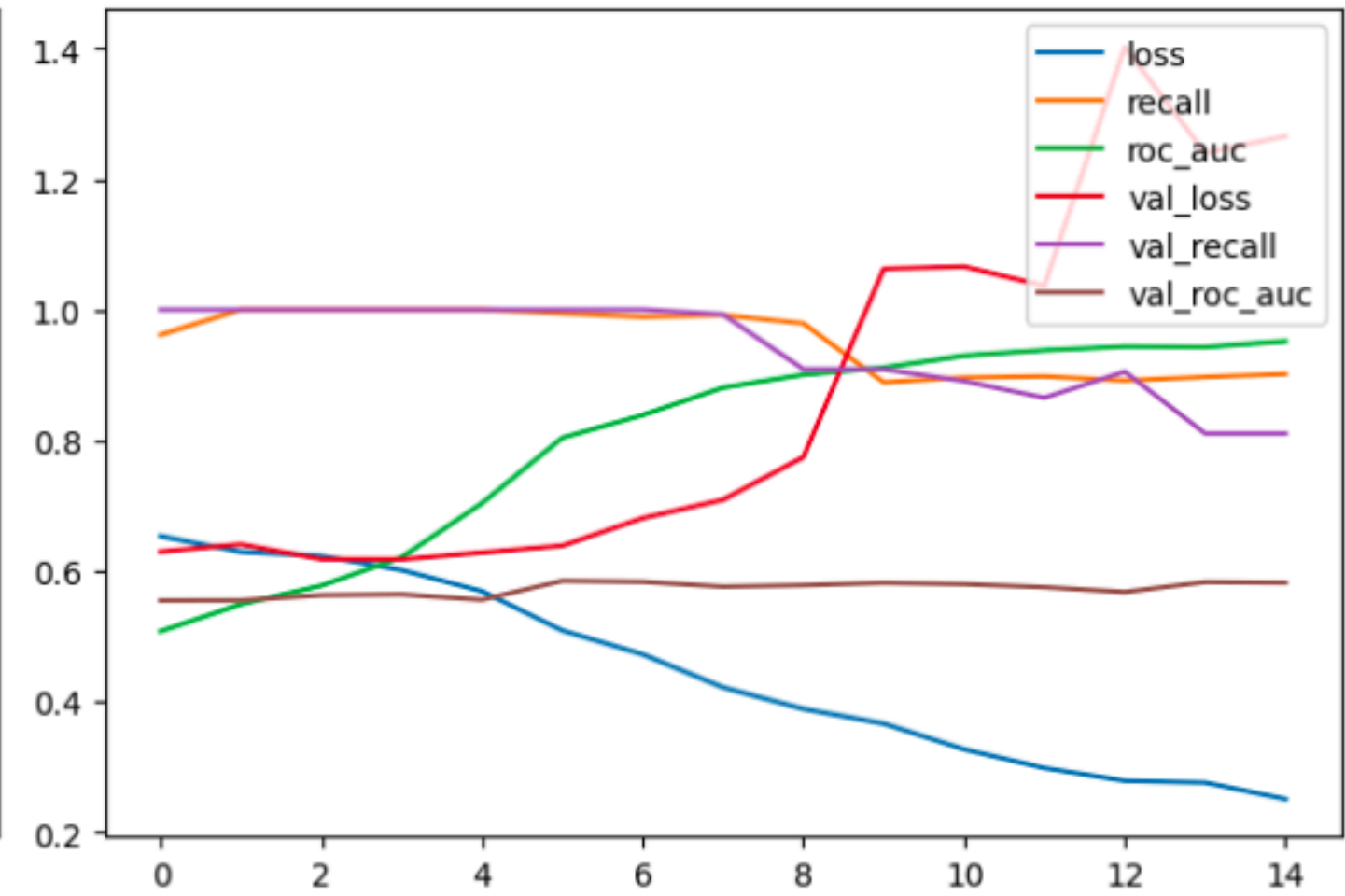


Models Benchmark

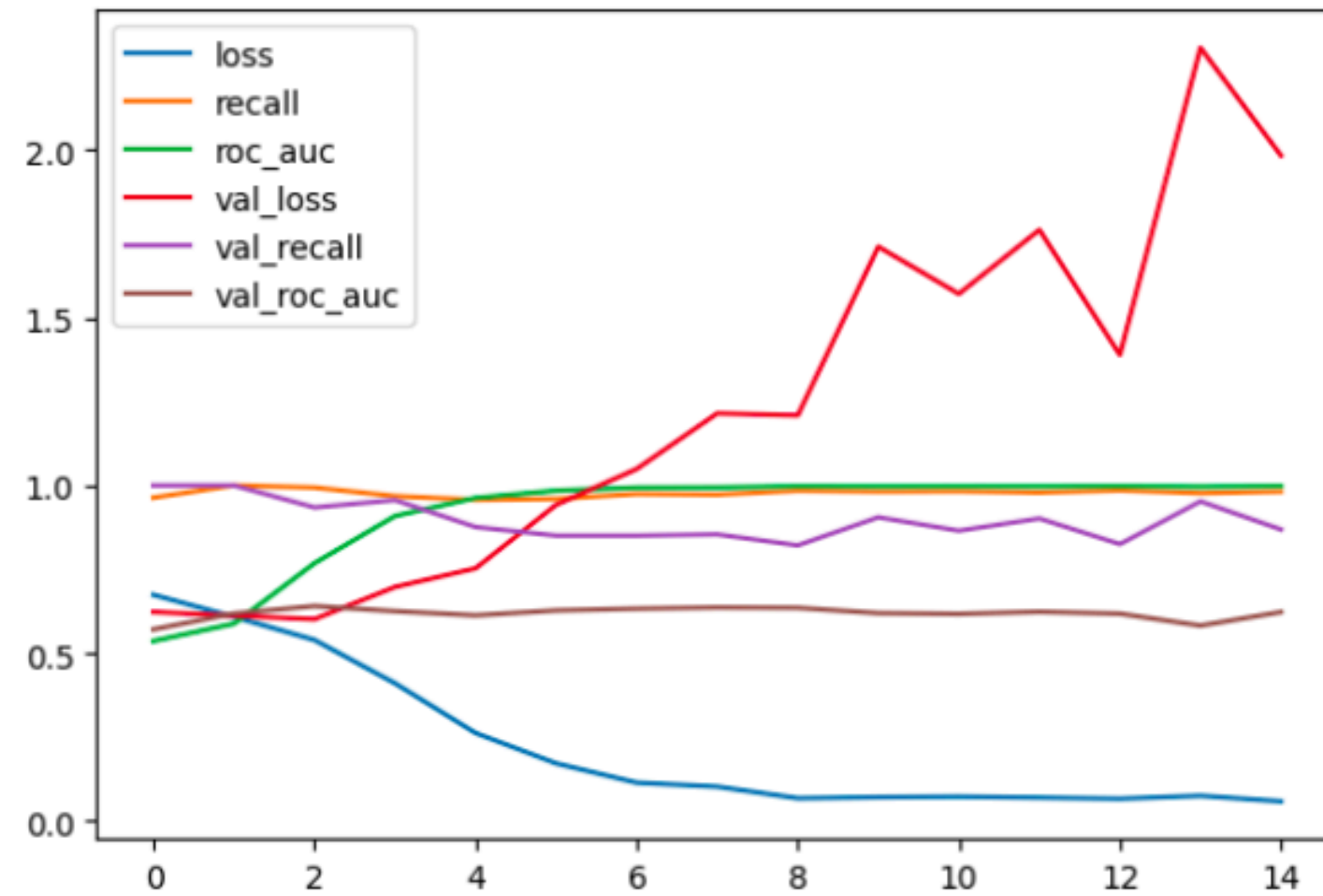
GPT proposed model



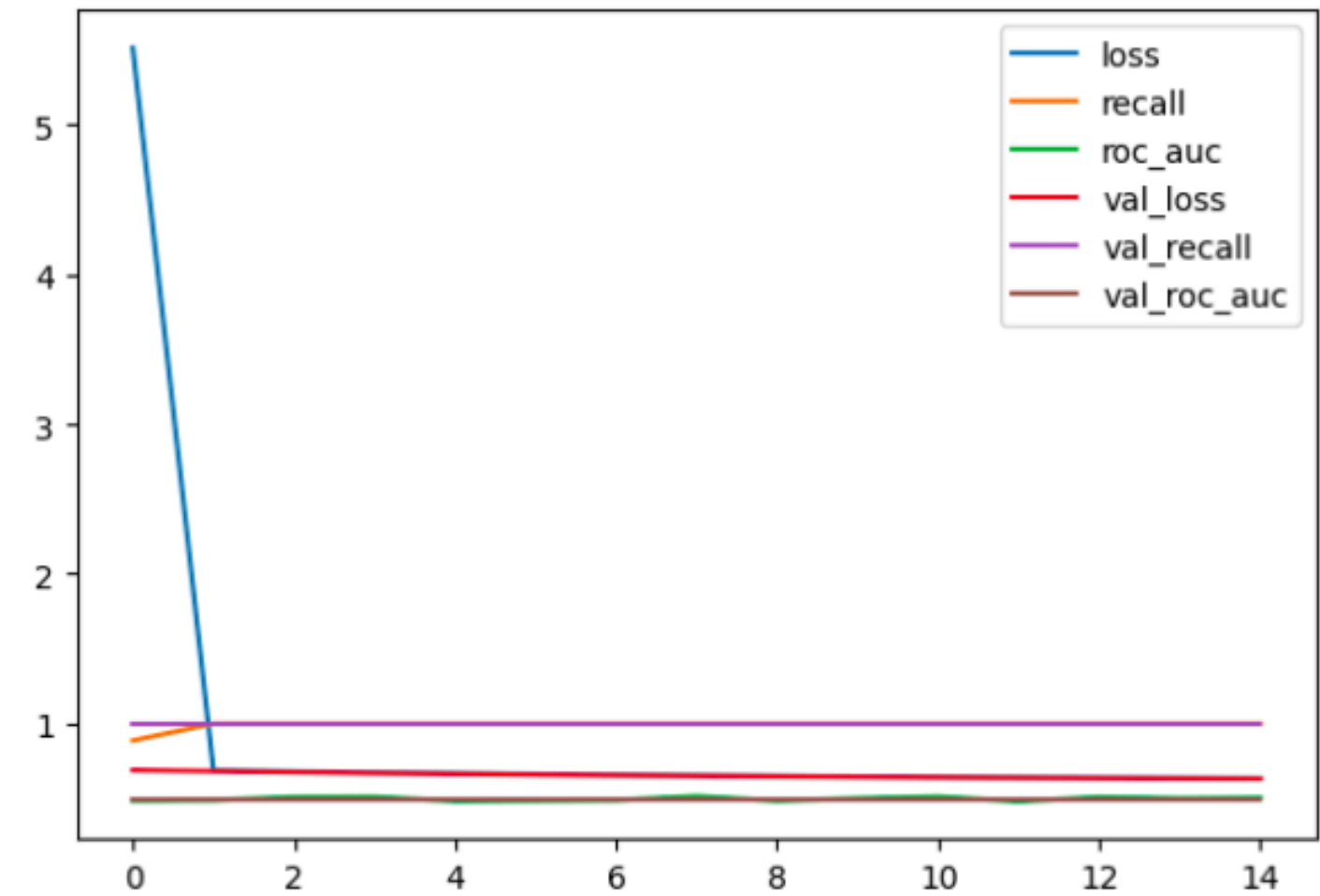
Refined model 1



Refined model 2

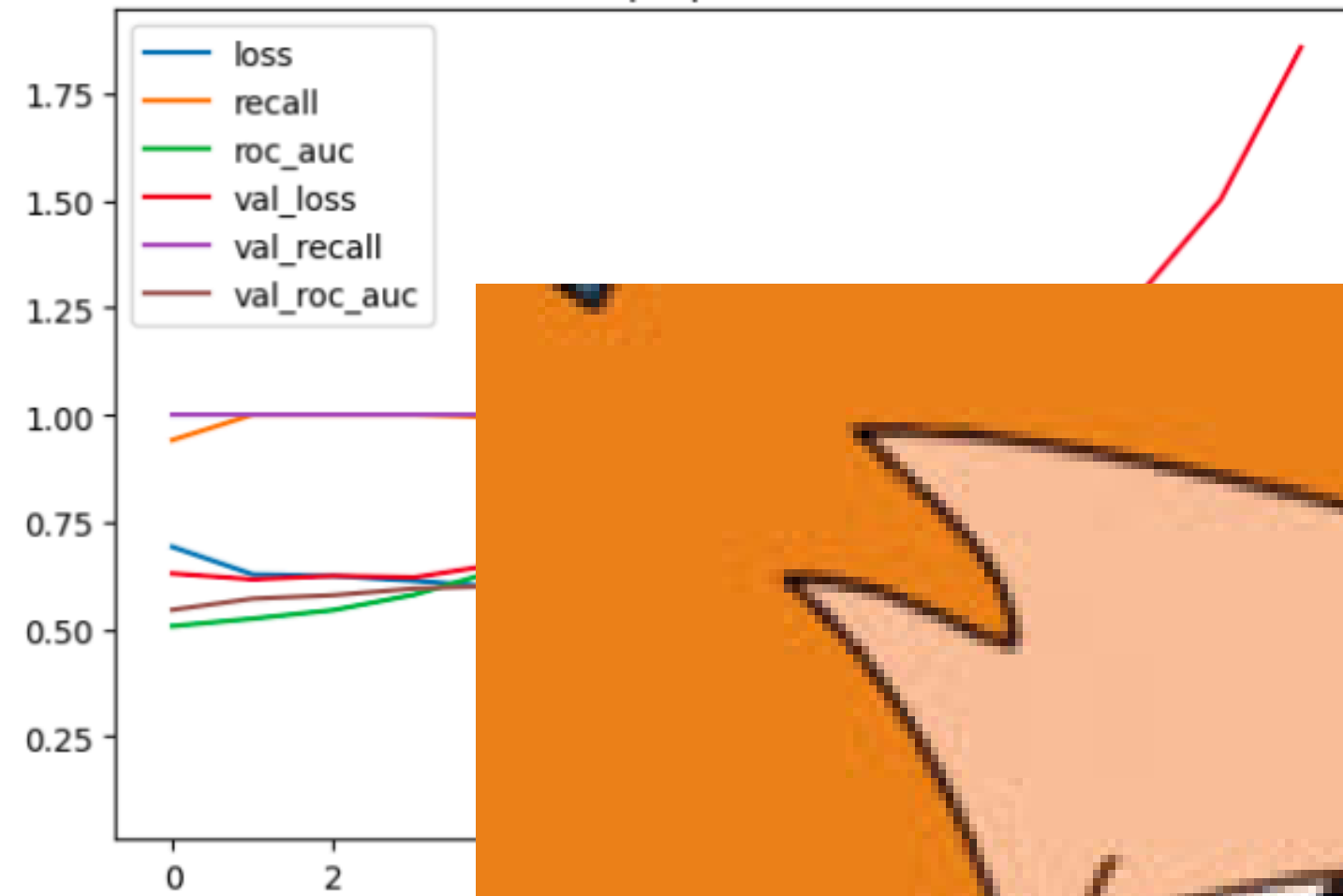


Refined model 3

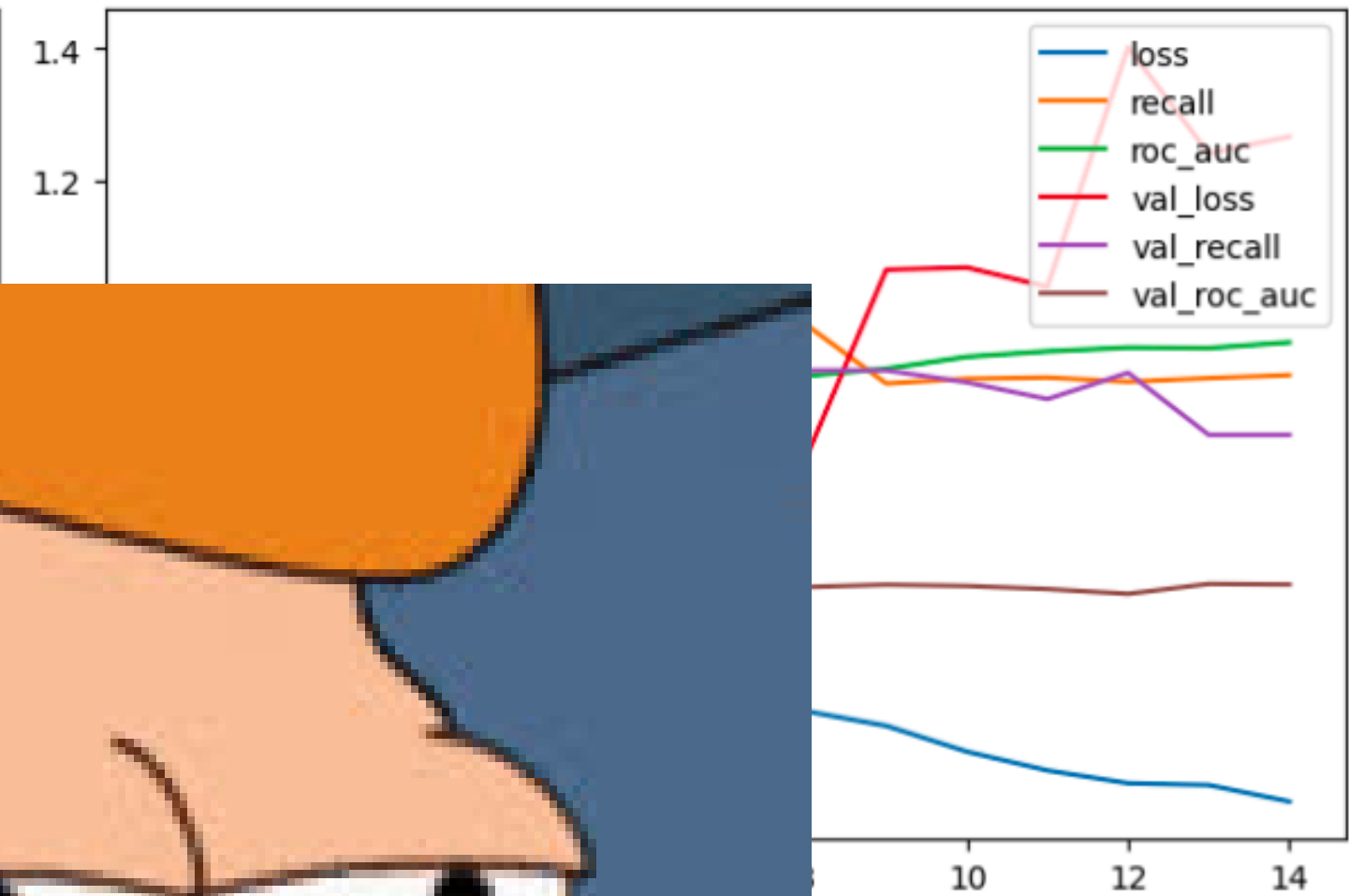


Models Benchmark

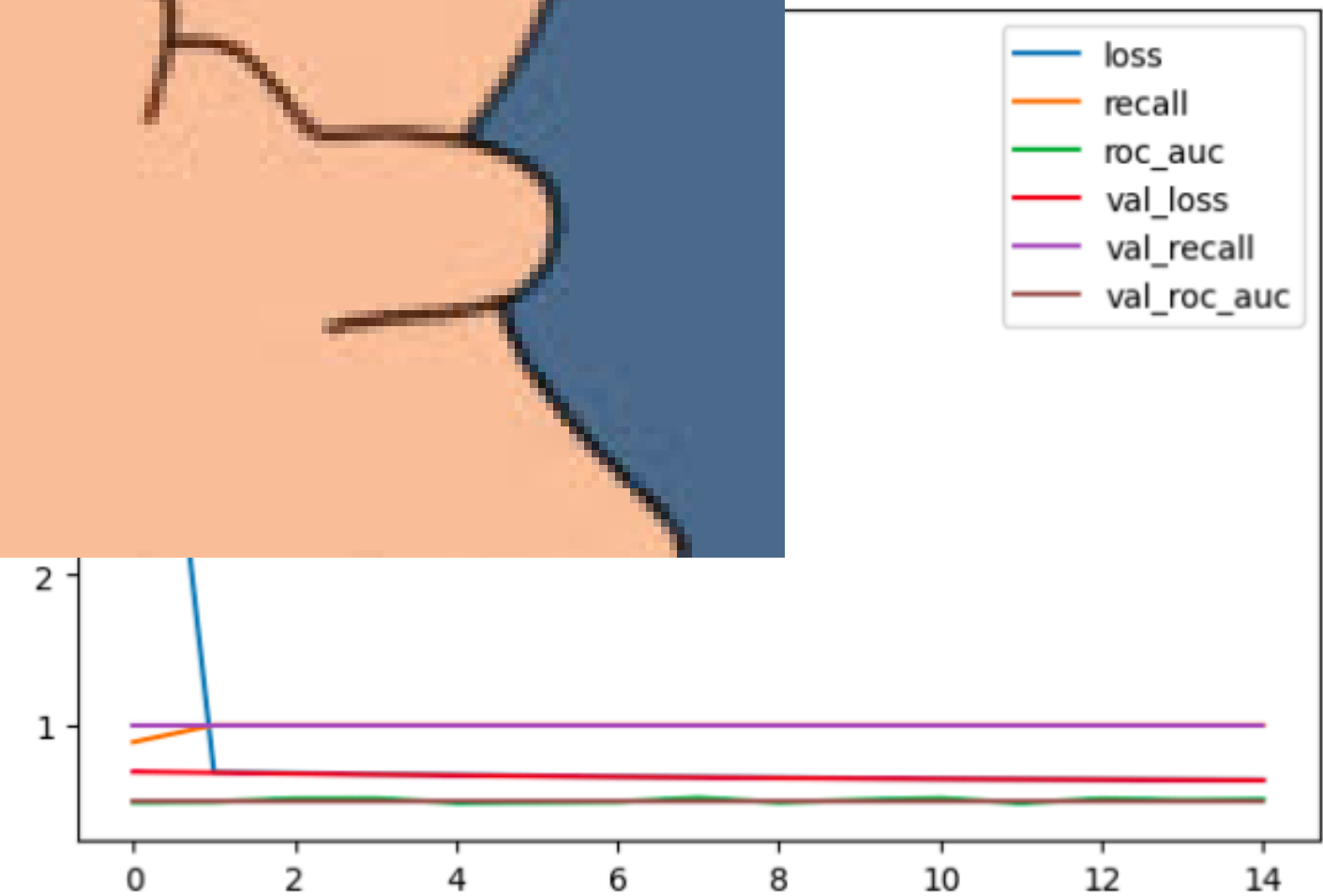
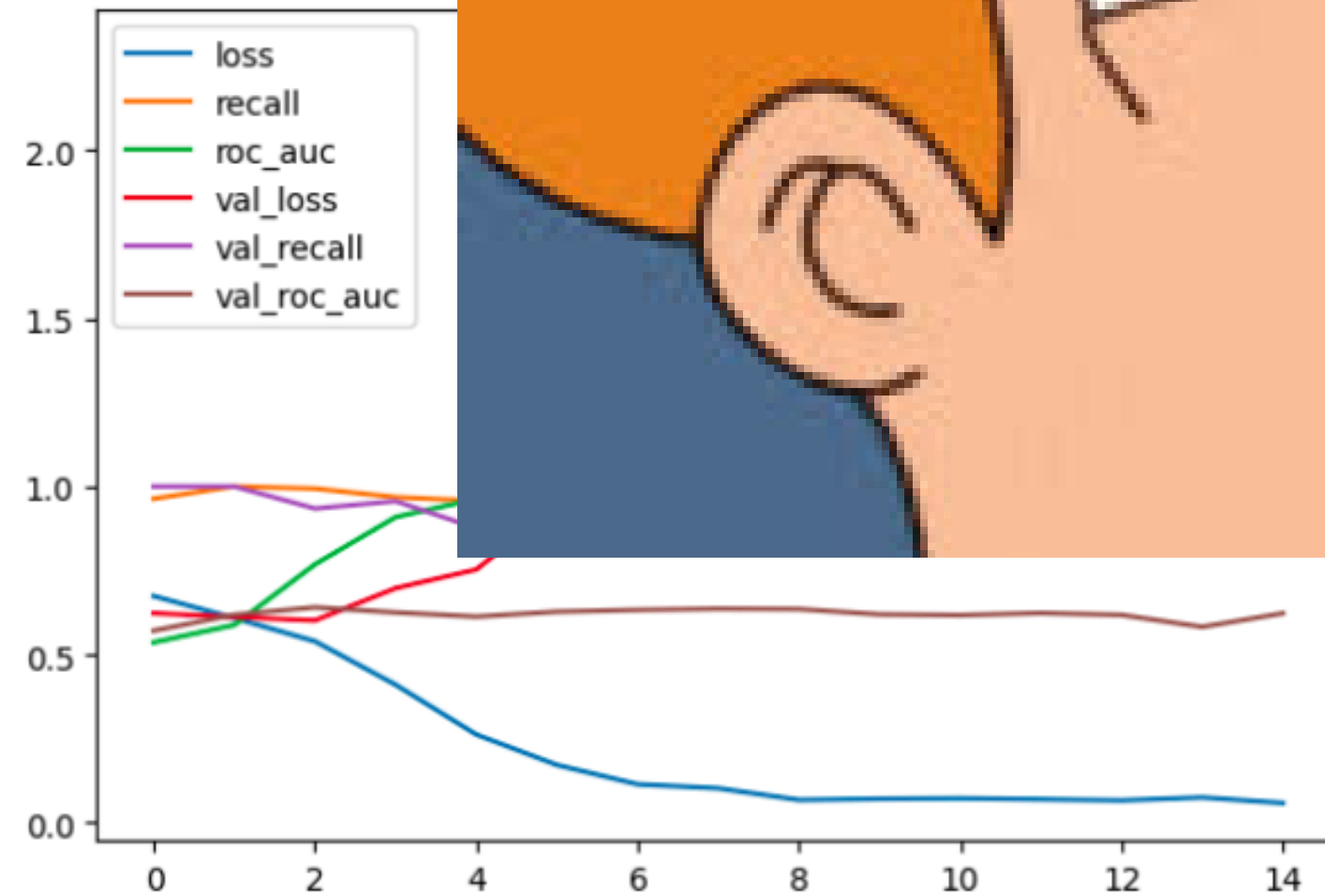
GPT proposed model



Refined model 1

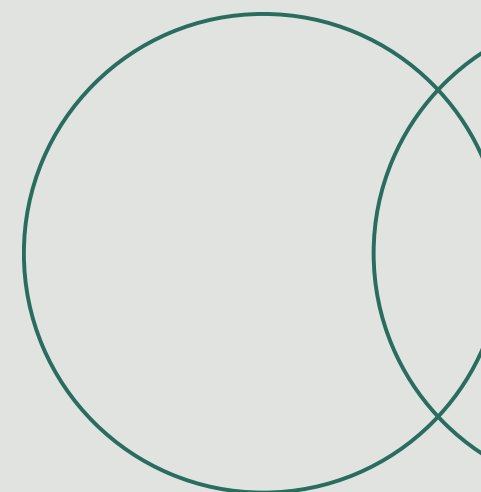
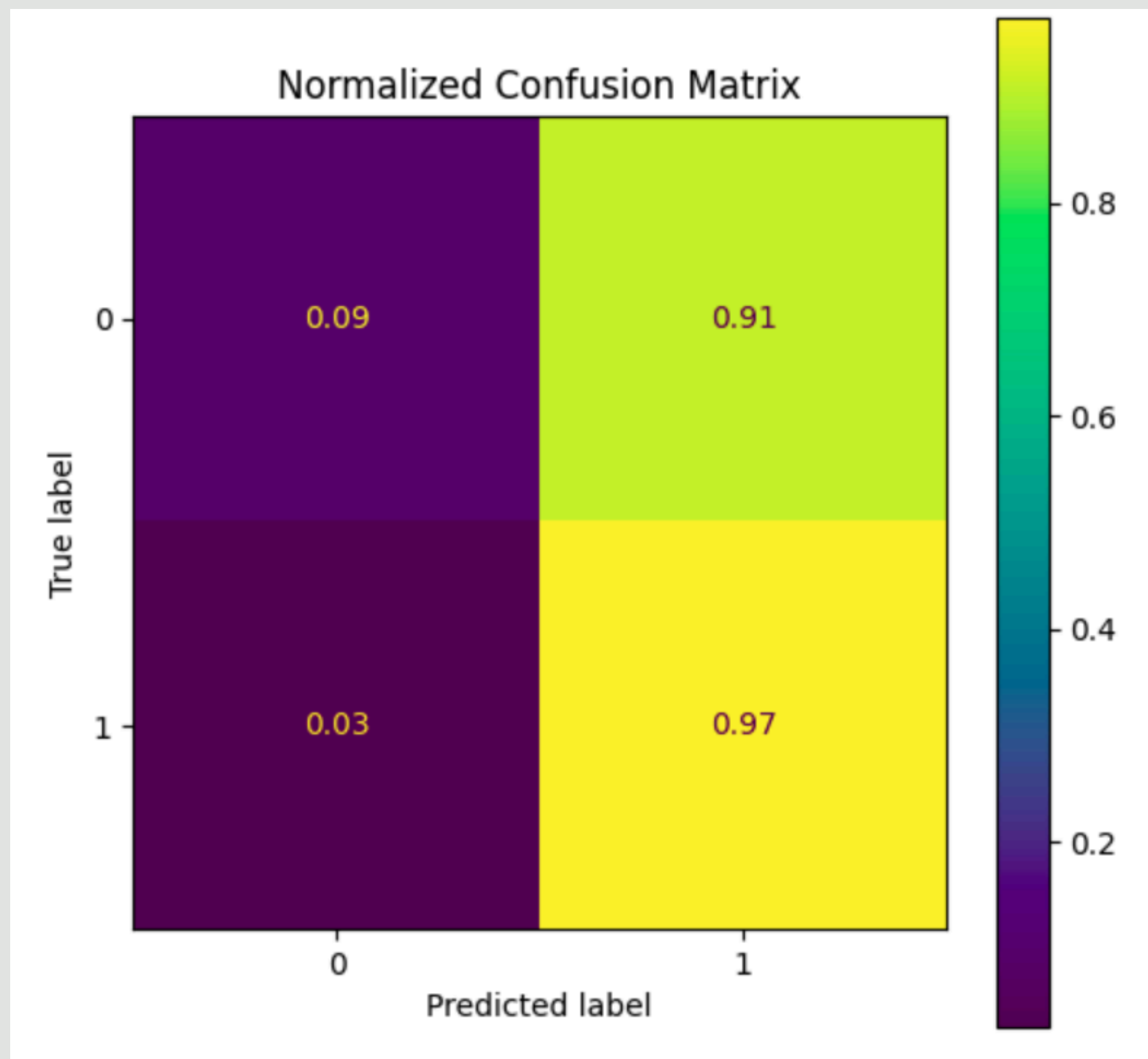


Model 3

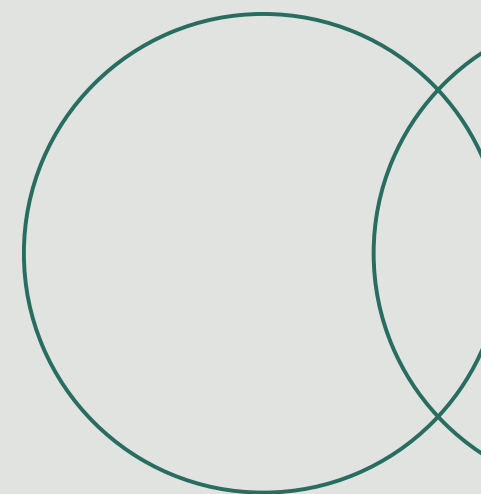
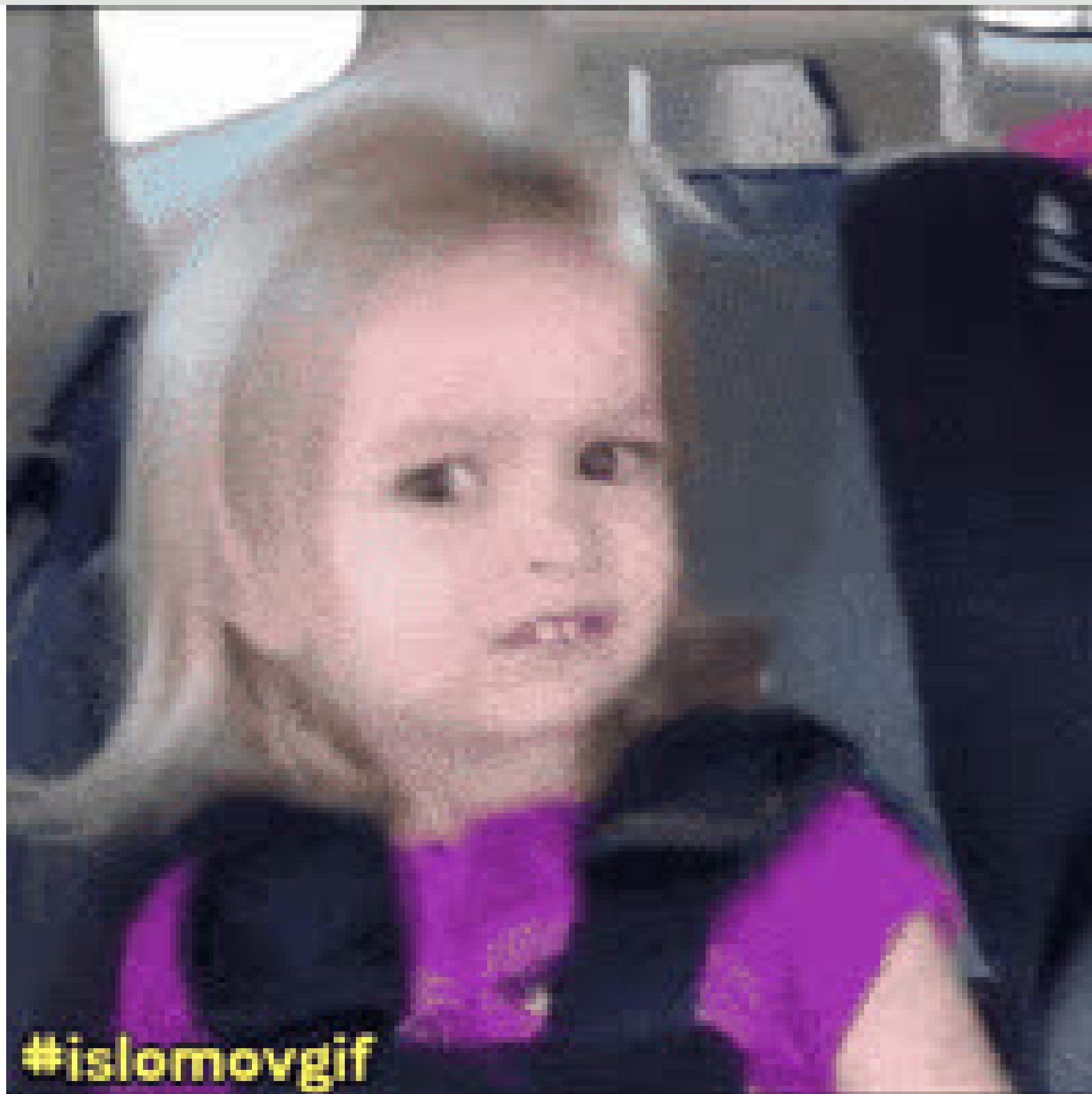




Mejor modelo:

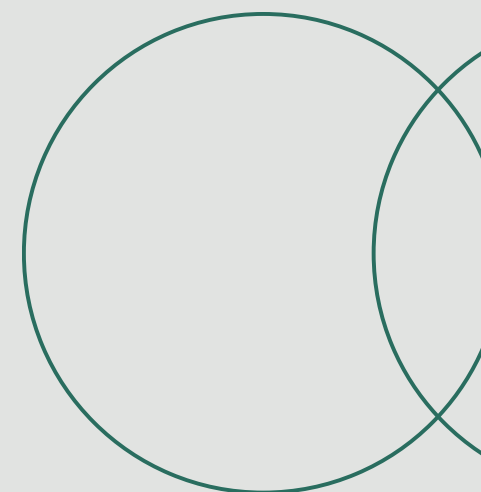


Mejor modelo:



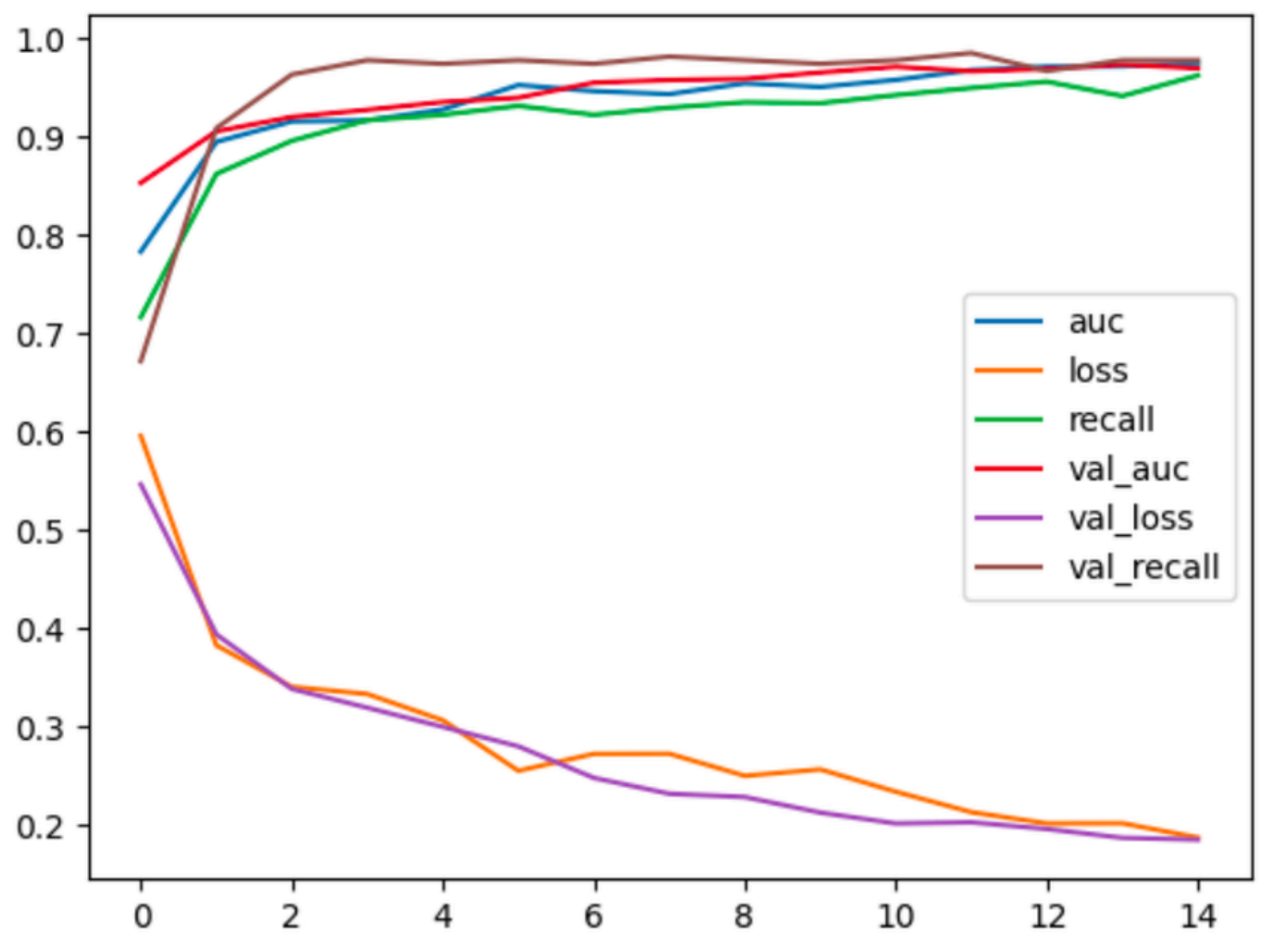
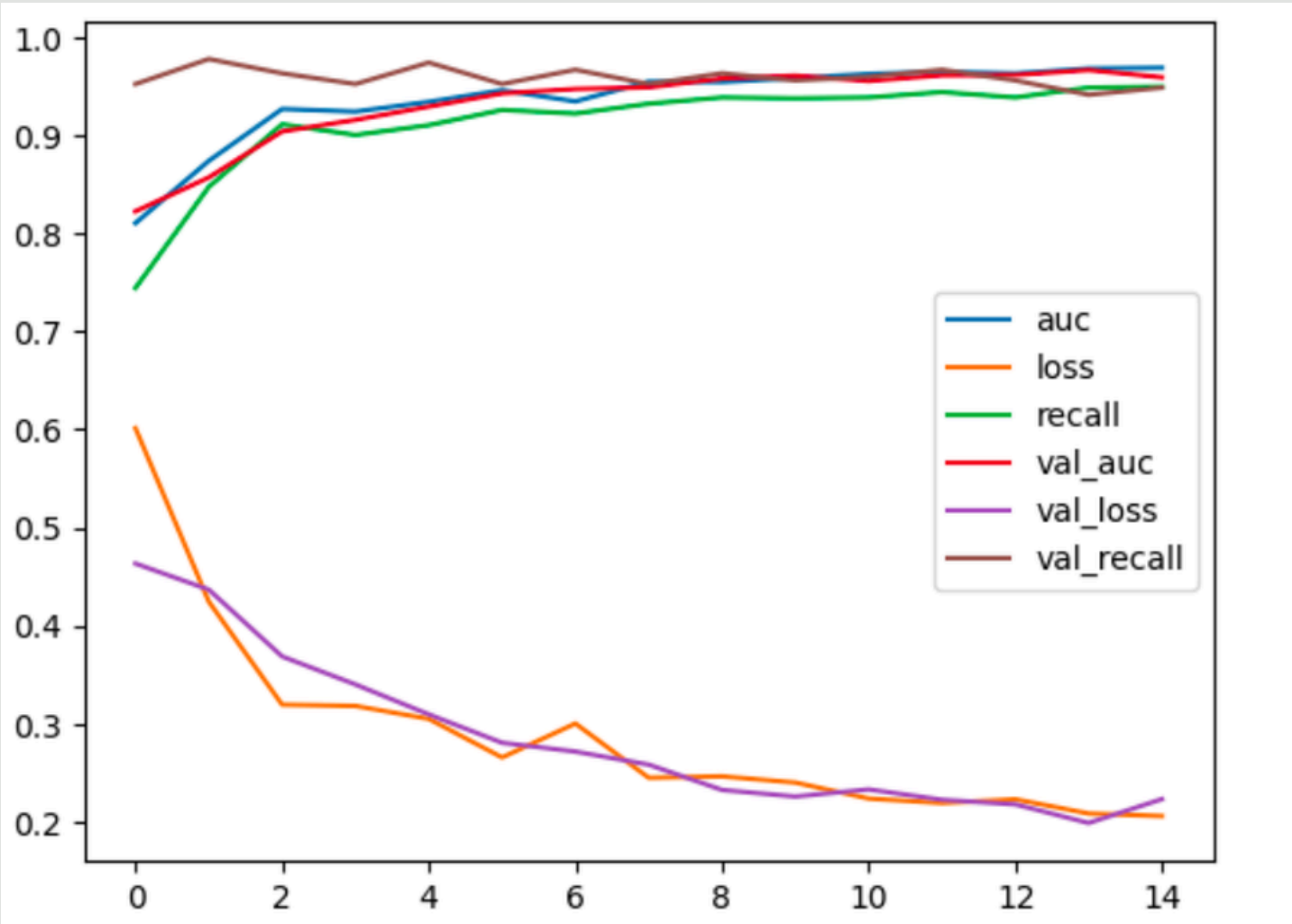


Inputs: Imagenes en escala gris



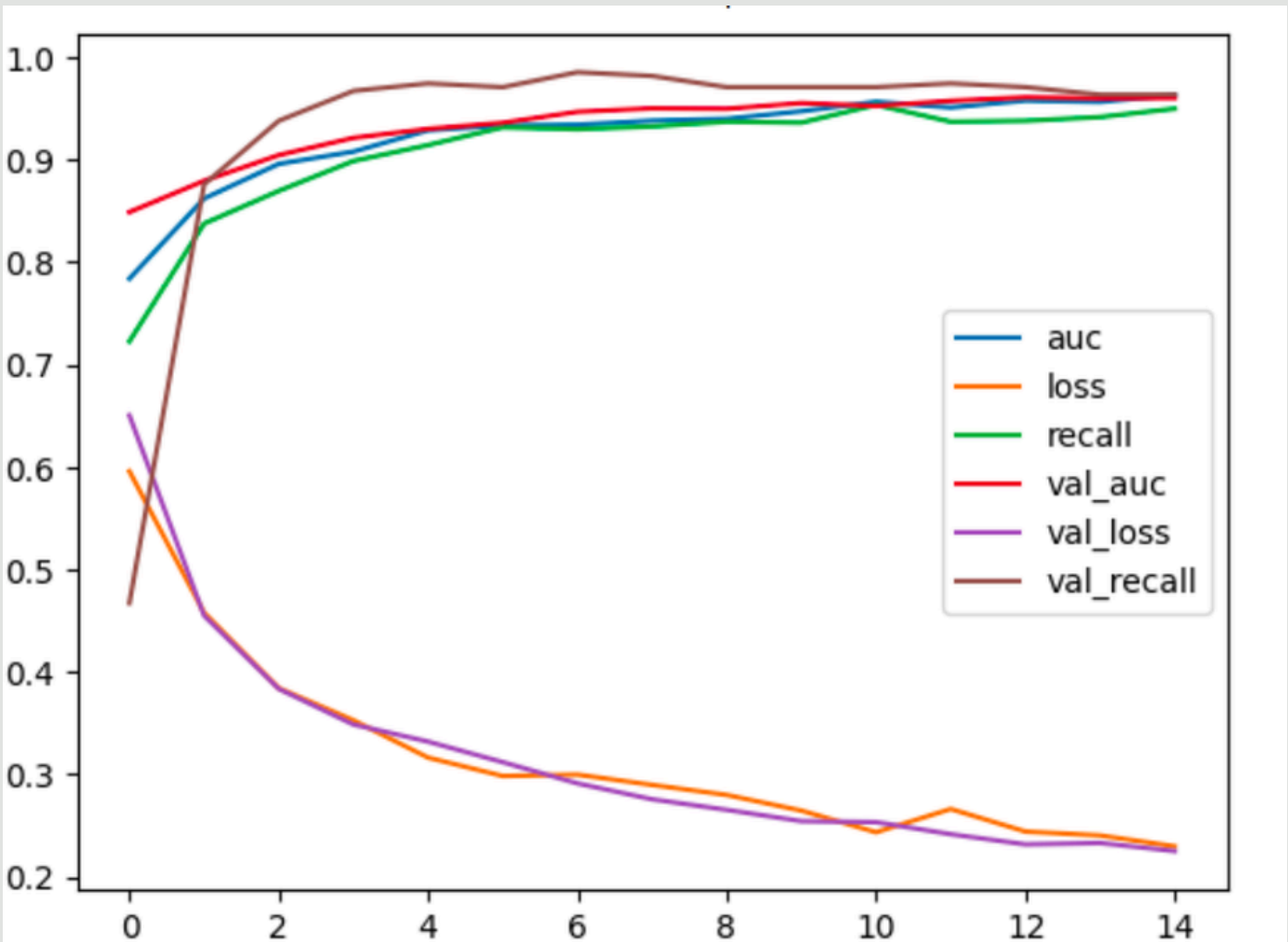
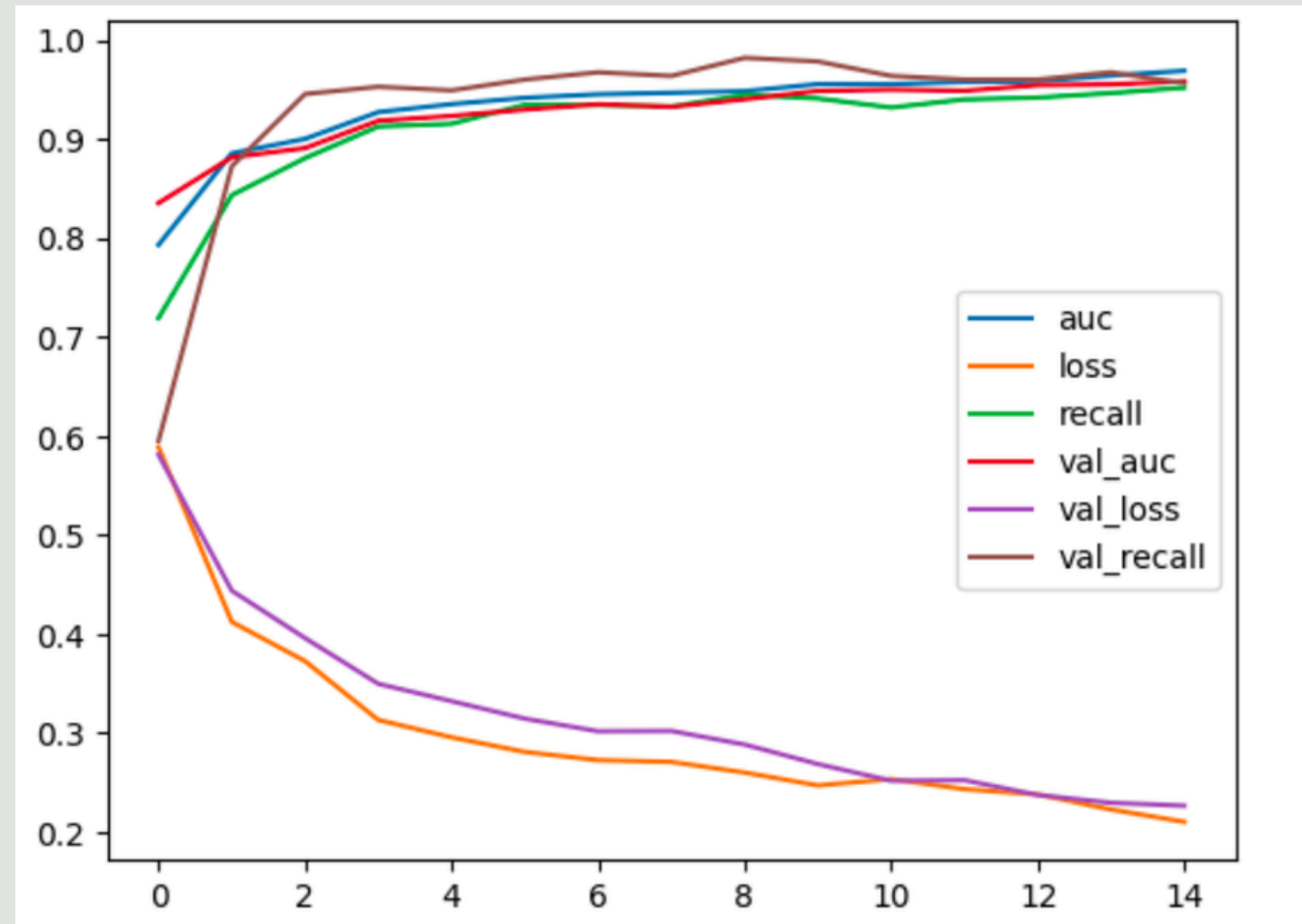


Modelo 0

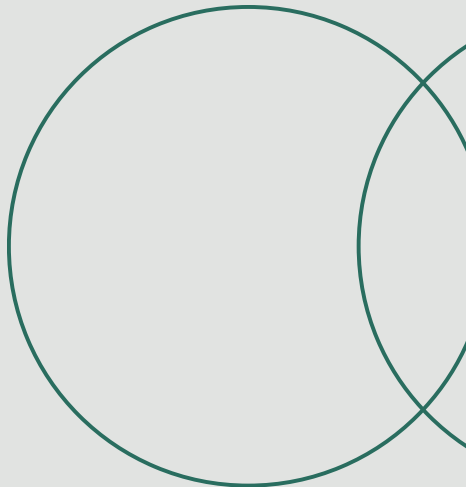


Modelo 1

Modelo 2

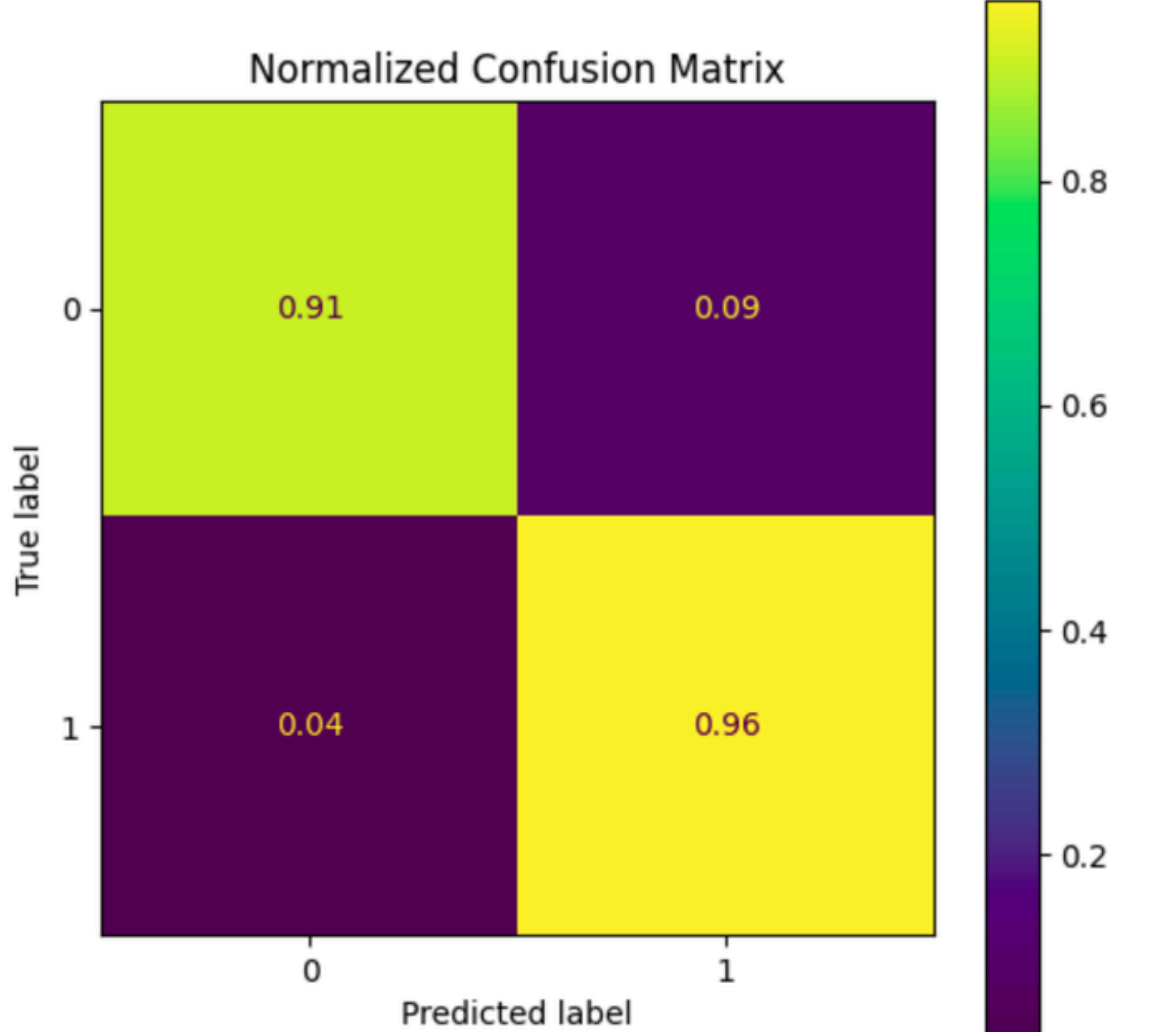
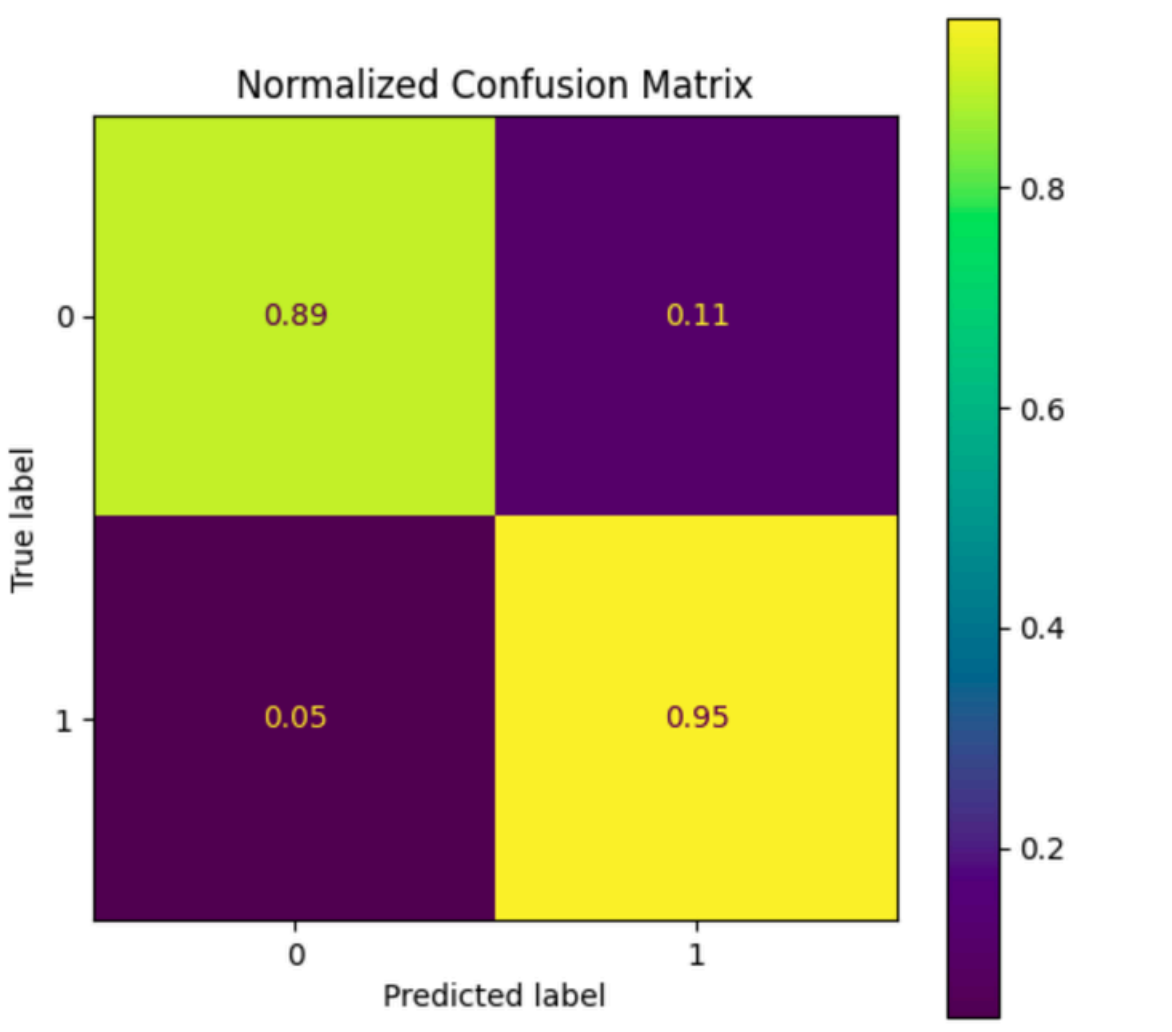


Modelo 3

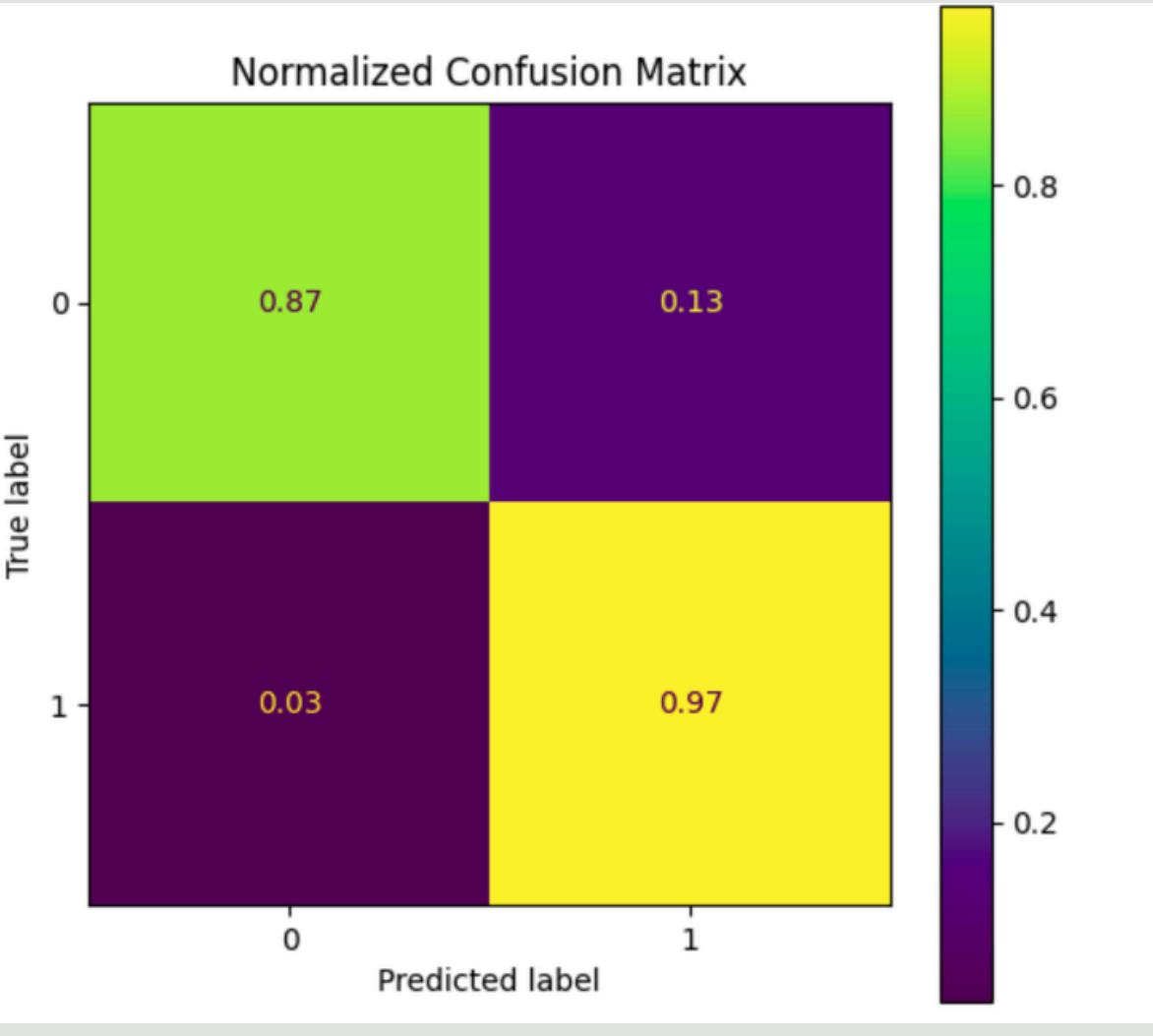




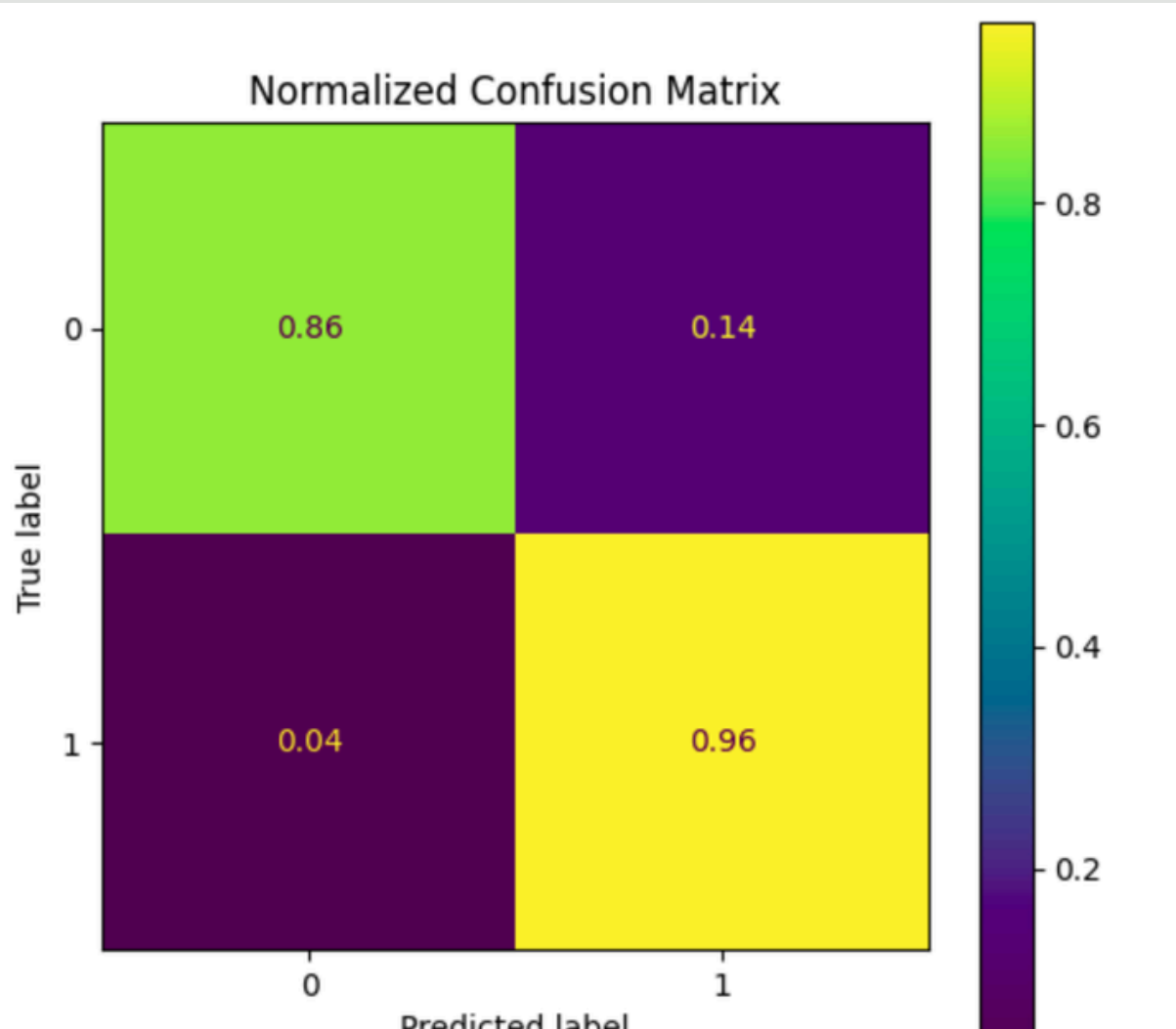
Modelo 0



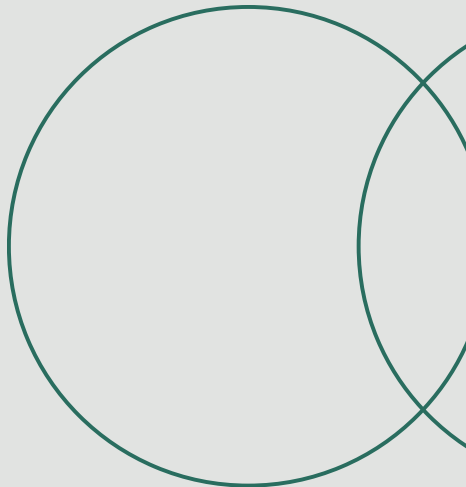
Modelo 1



Modelo 2



Modelo 3

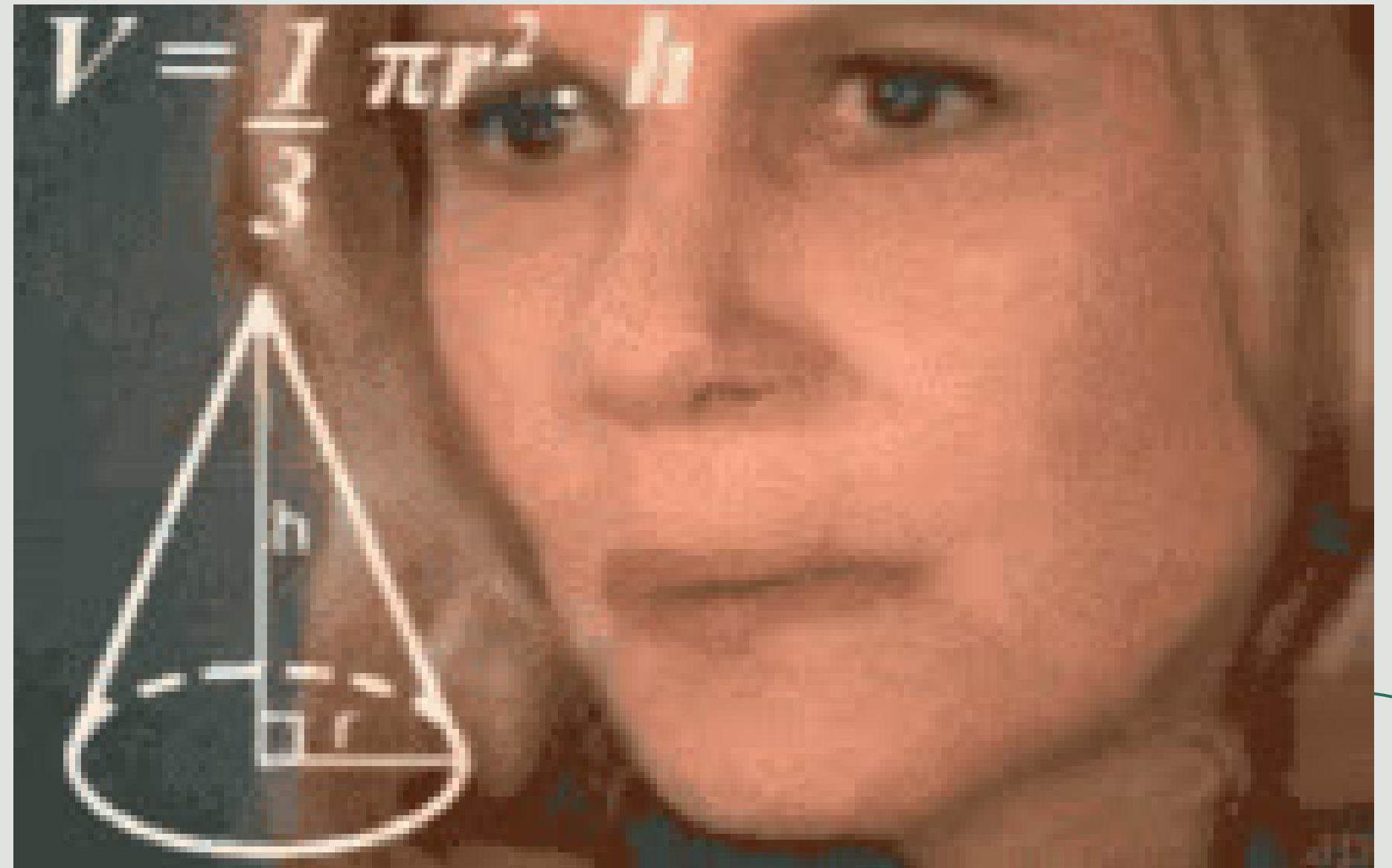




¿Y si incluyo un balanceo de clases en la información de entrenamiento?

¿Qué pesos debo atribuir a cada clase?

¿Más peso a los benignos o malignos?



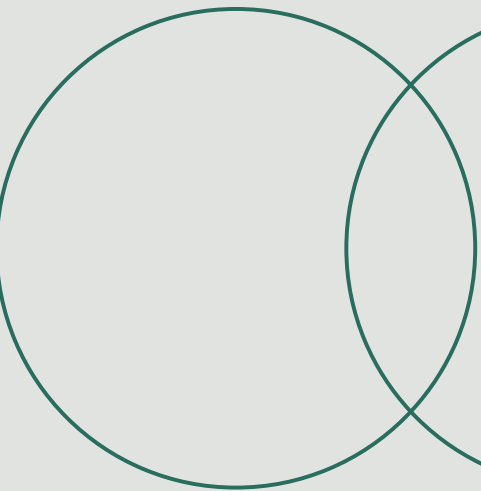


Cálculo de pesos:

$$w_c = \frac{N}{2n_c}$$

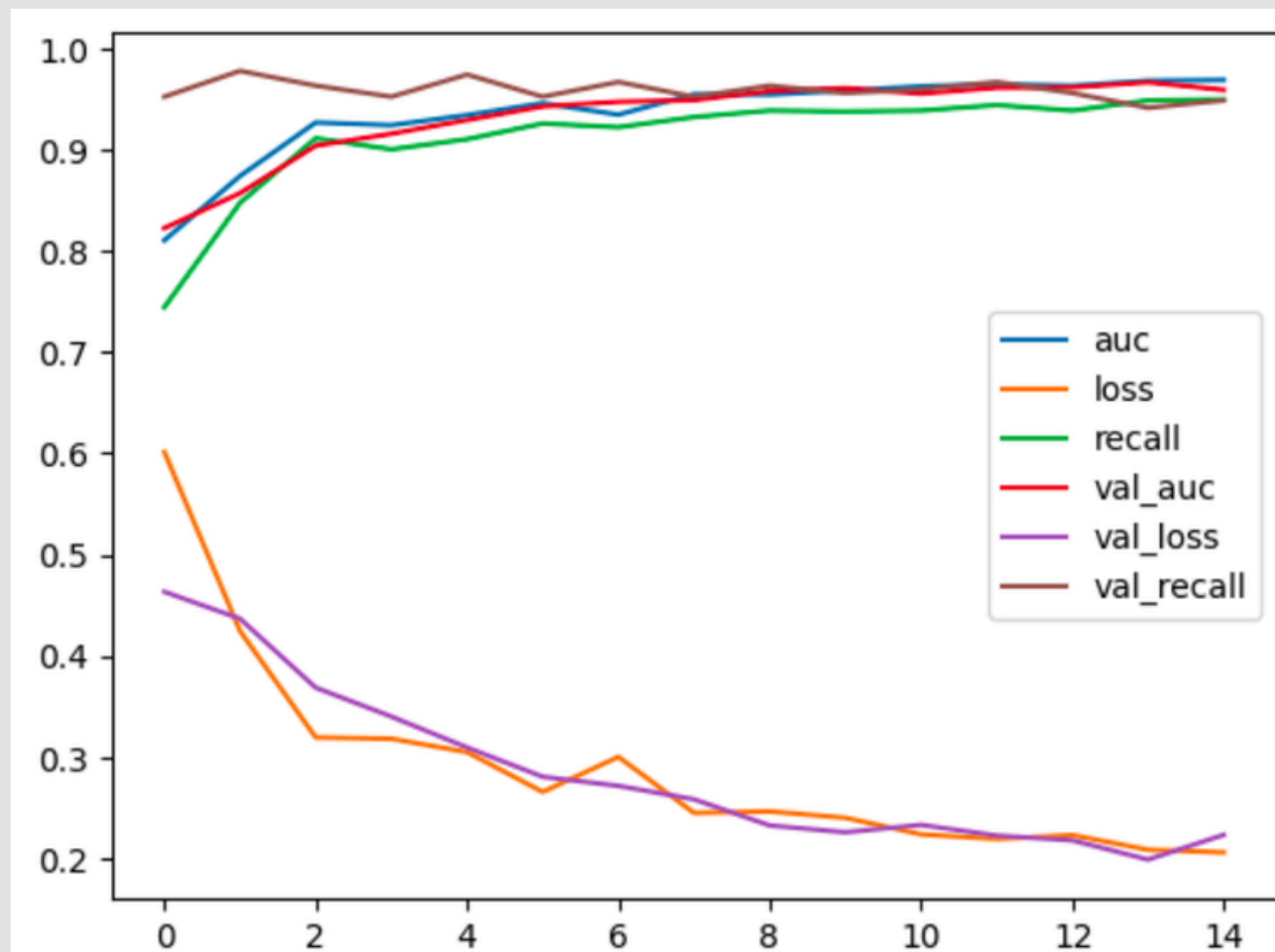
N: número total de imágenes

n: Número de imágenes en la clase c

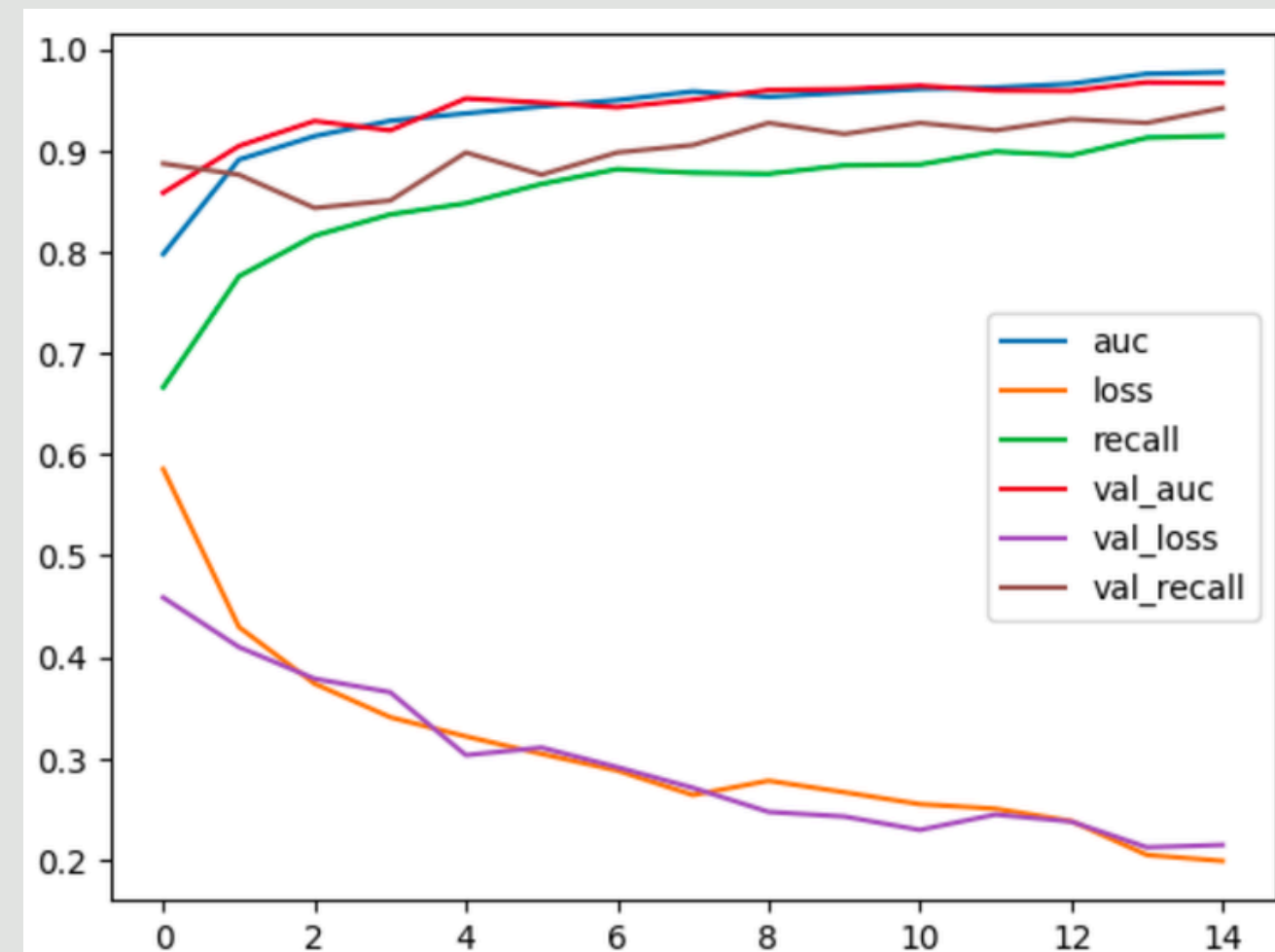




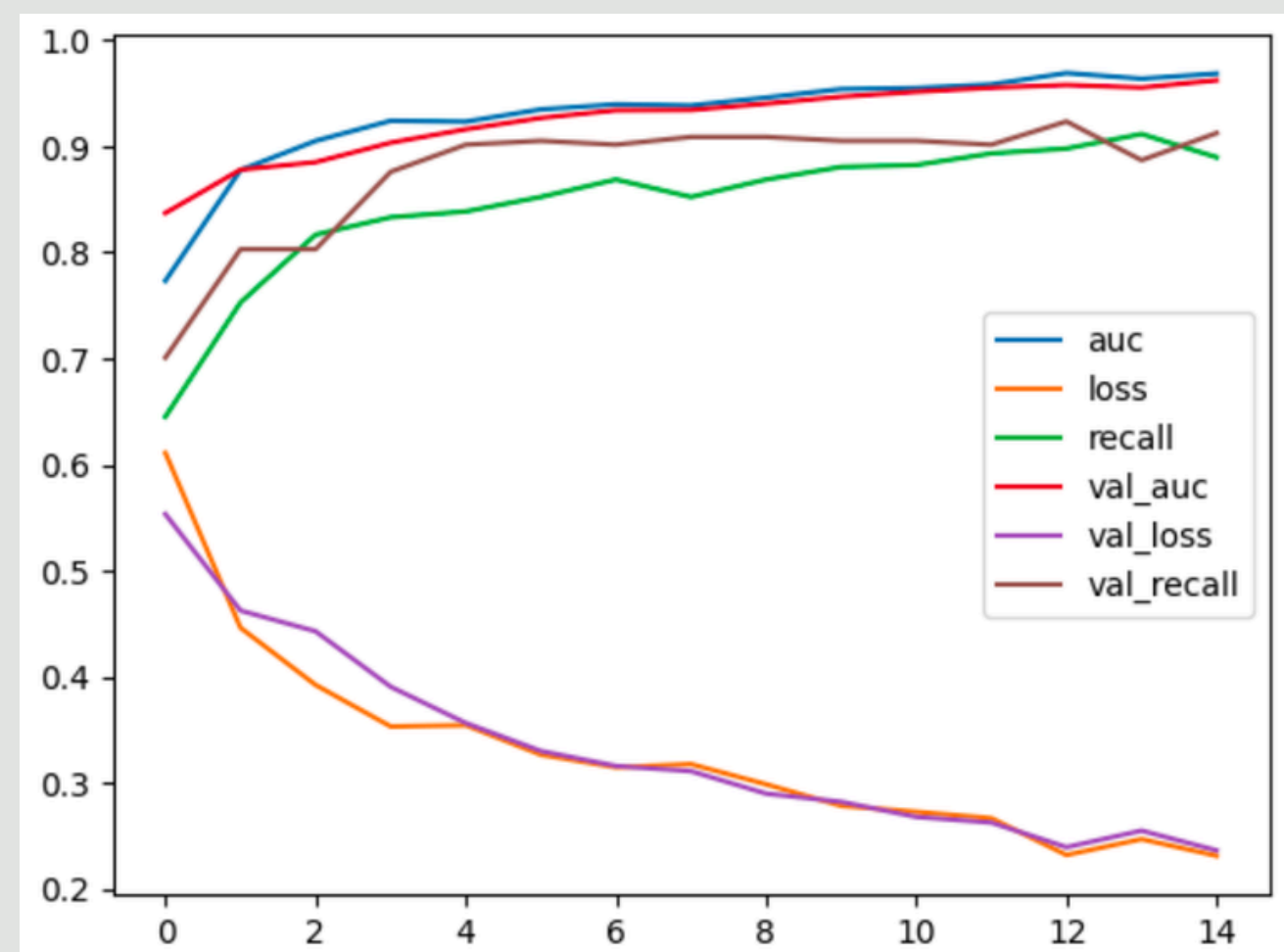
Modelo 0



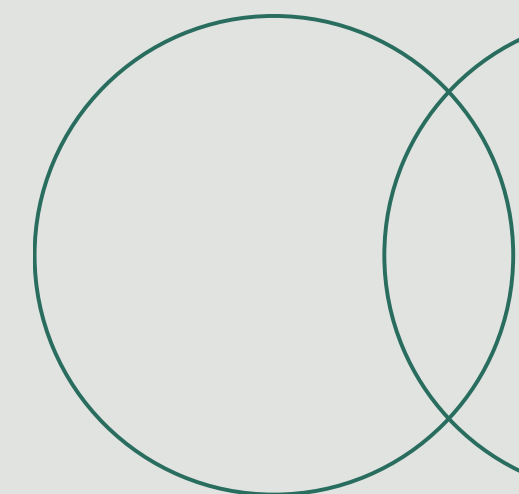
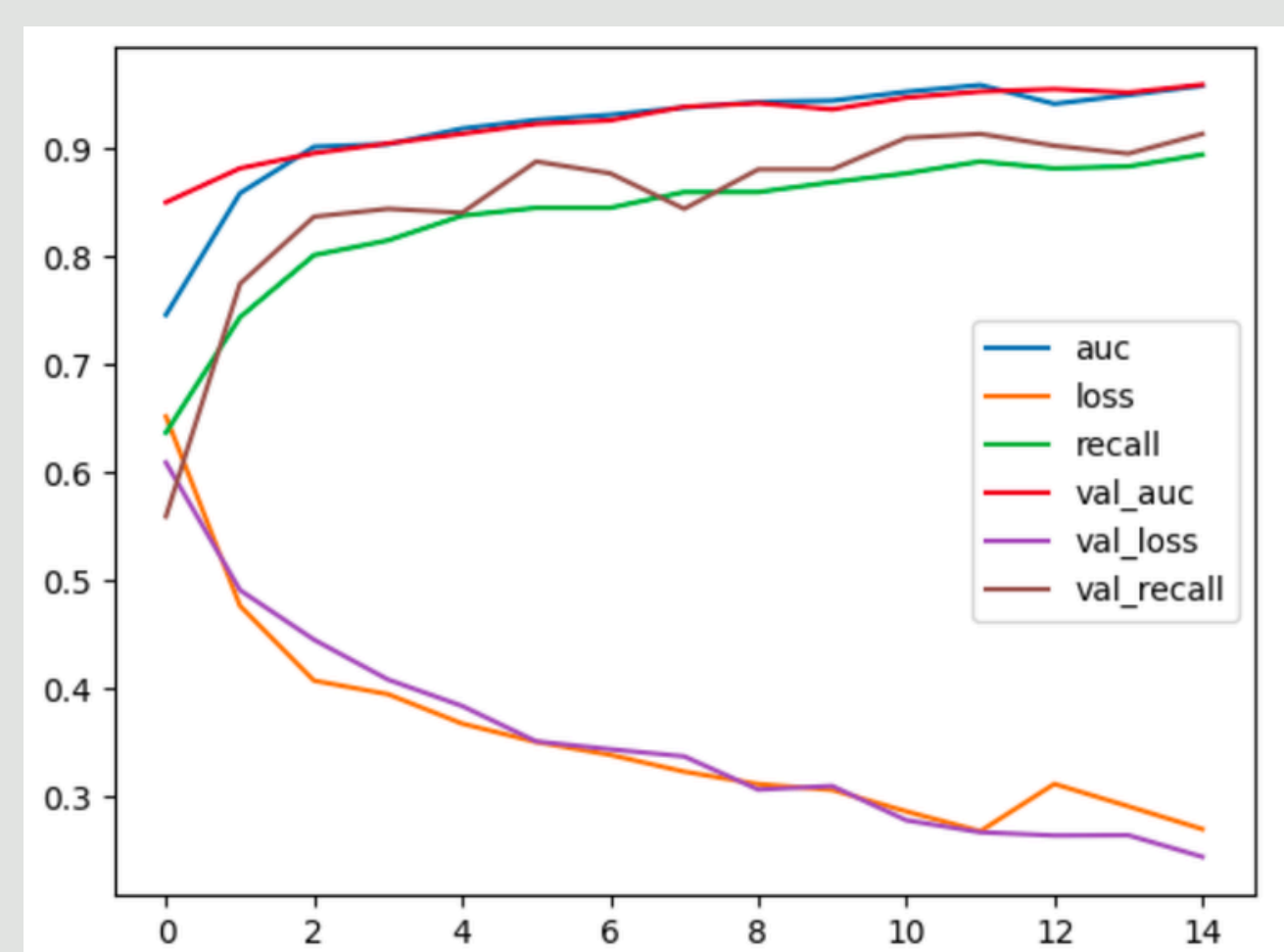
Modelo 1

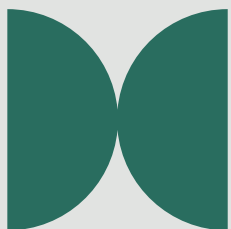


Modelo 2

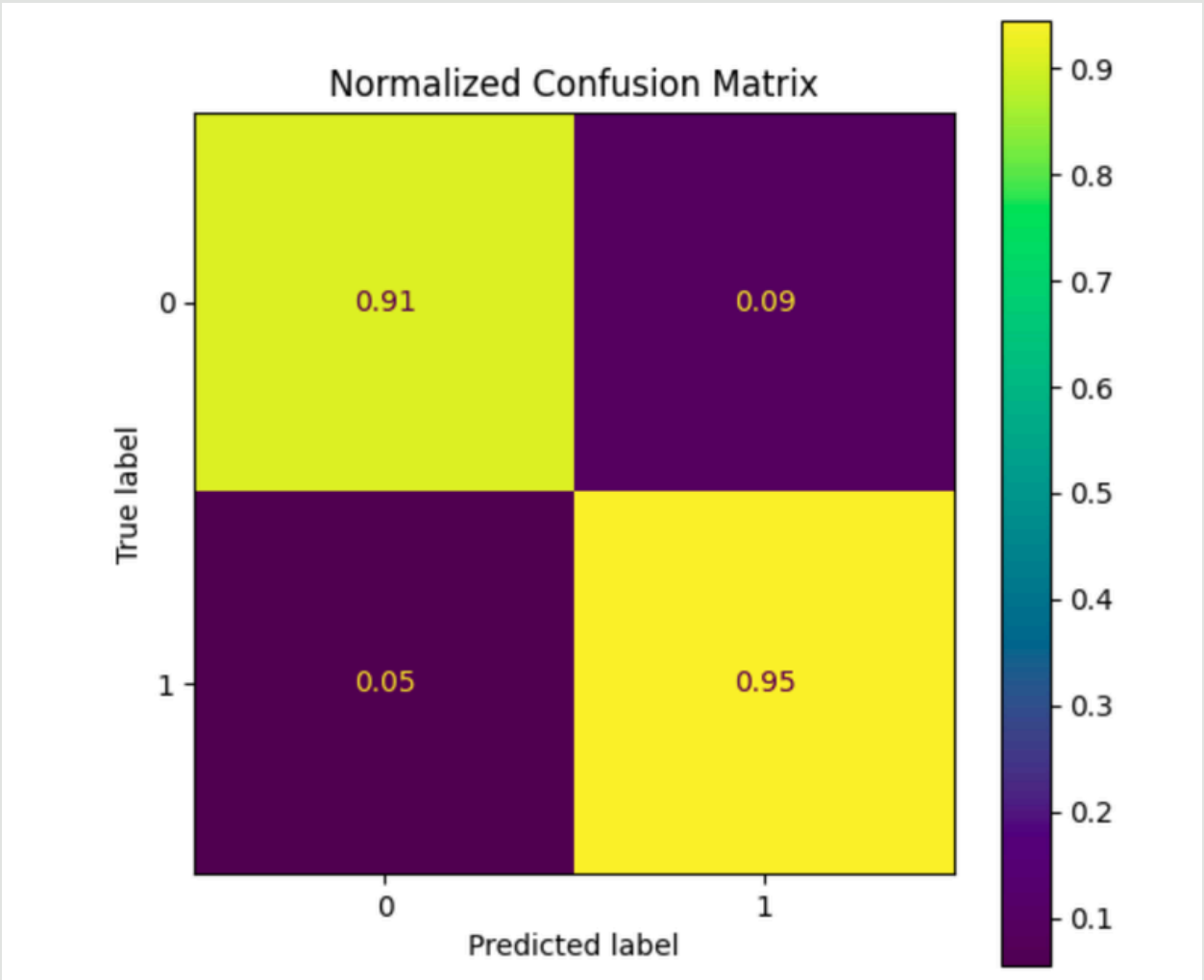
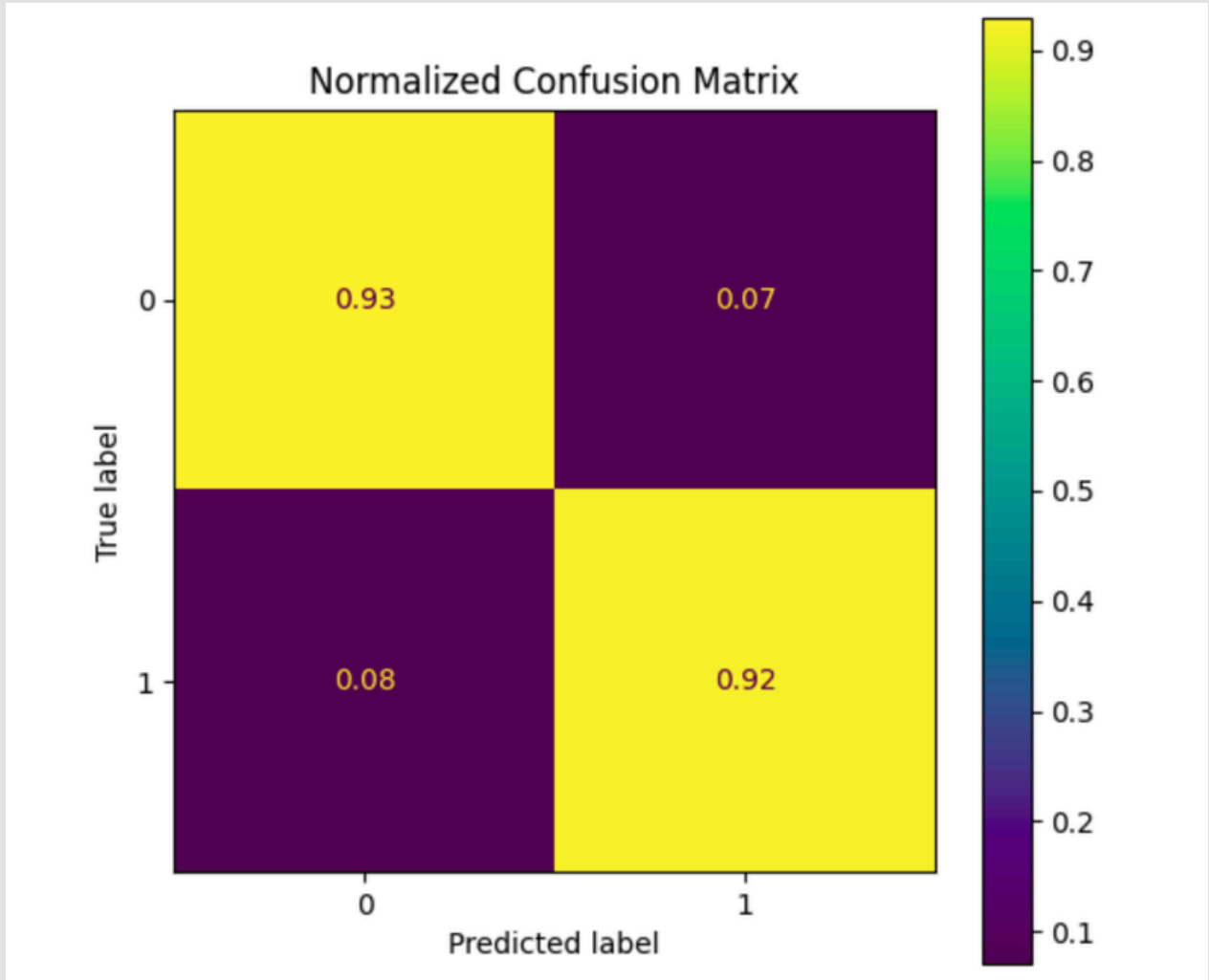


Modelo 3



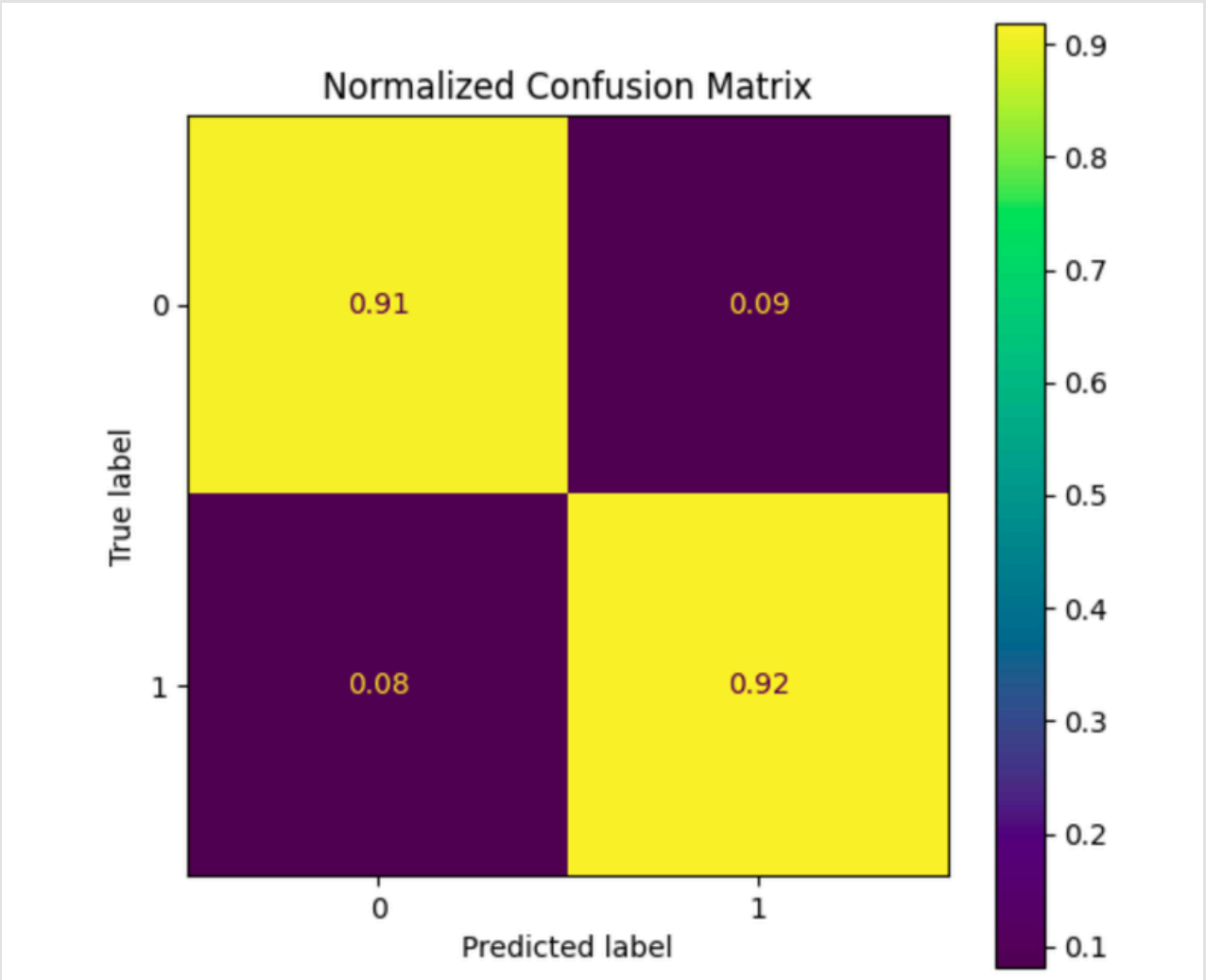
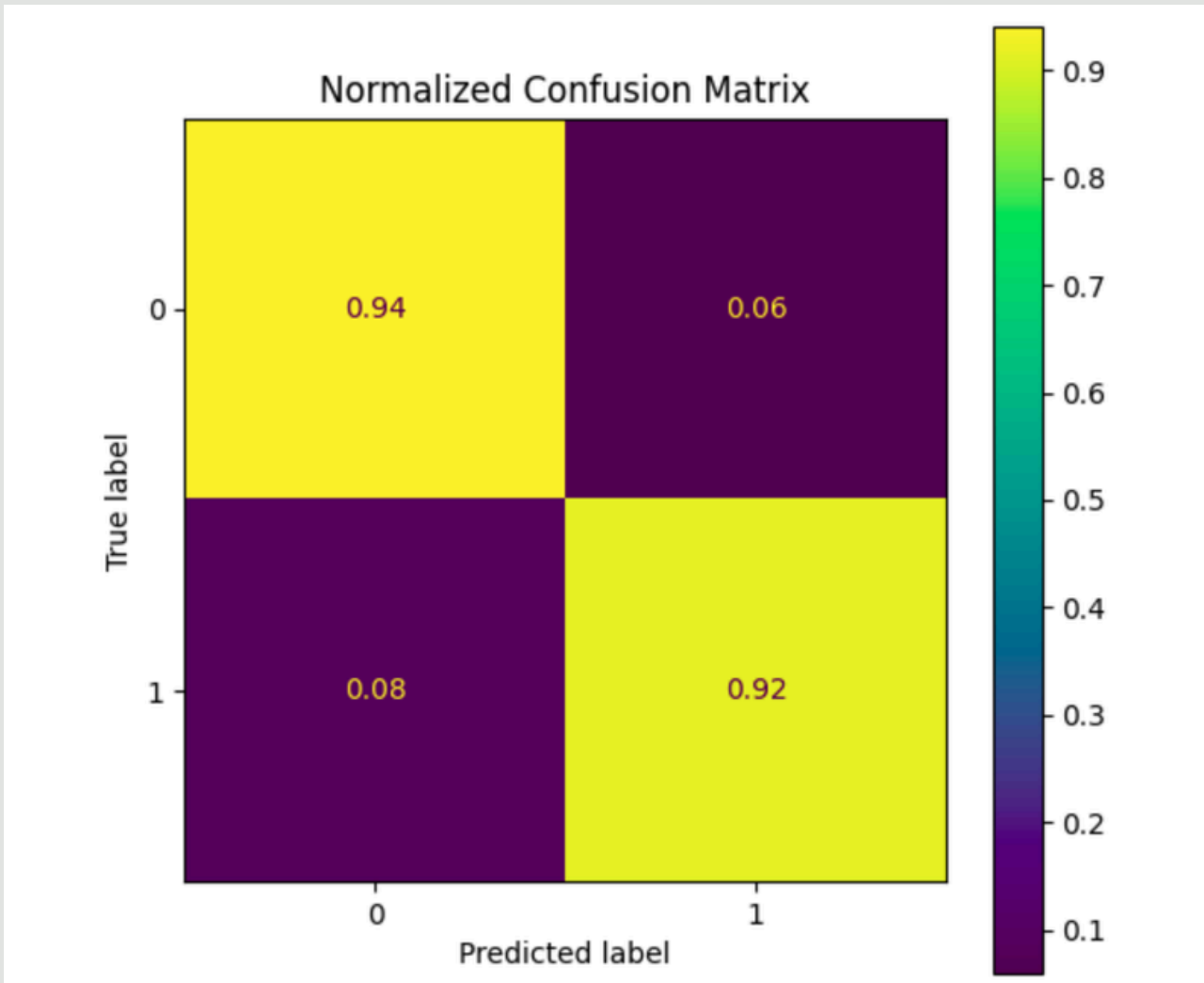


Modelo 0

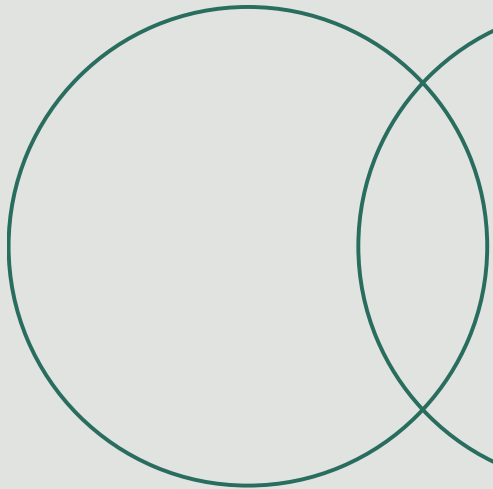


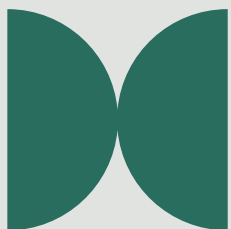
Modelo 1

Modelo 2

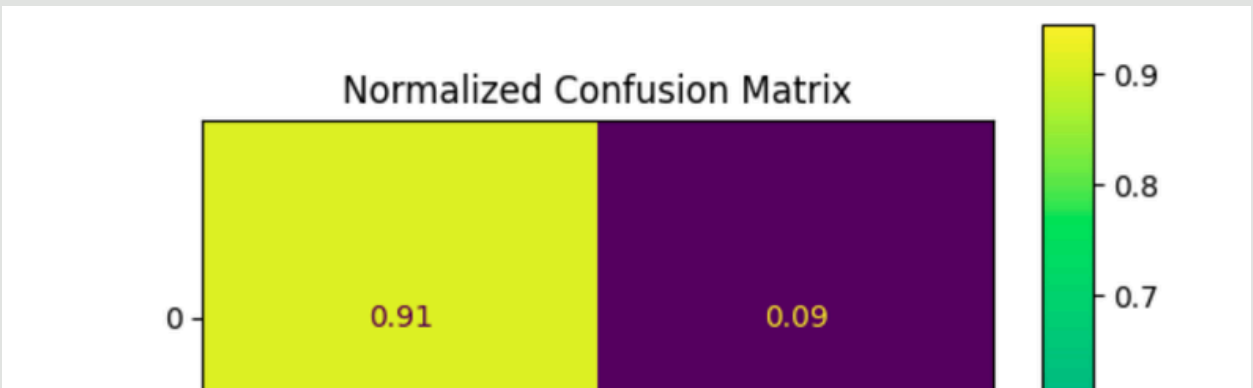
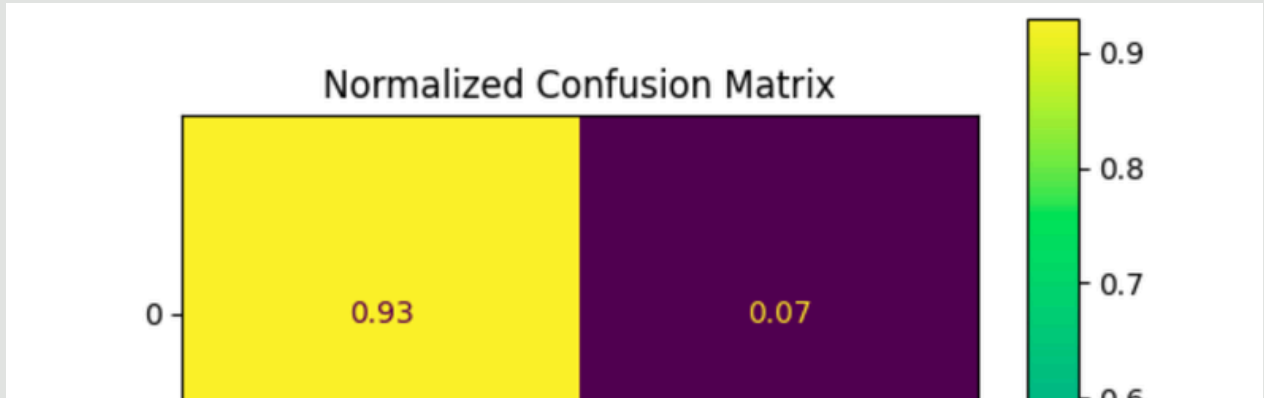


Modelo 3

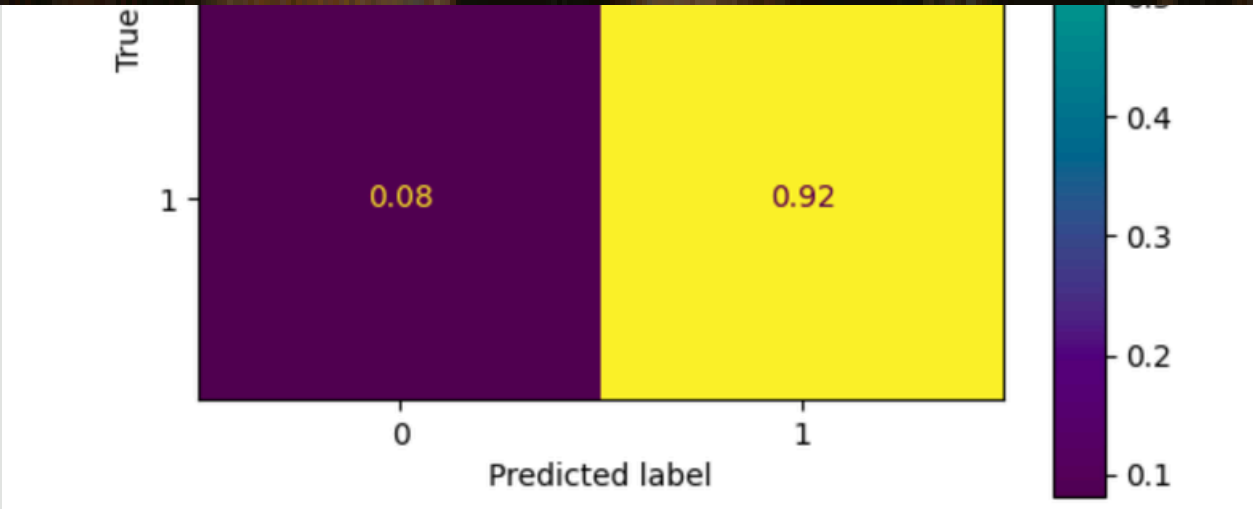
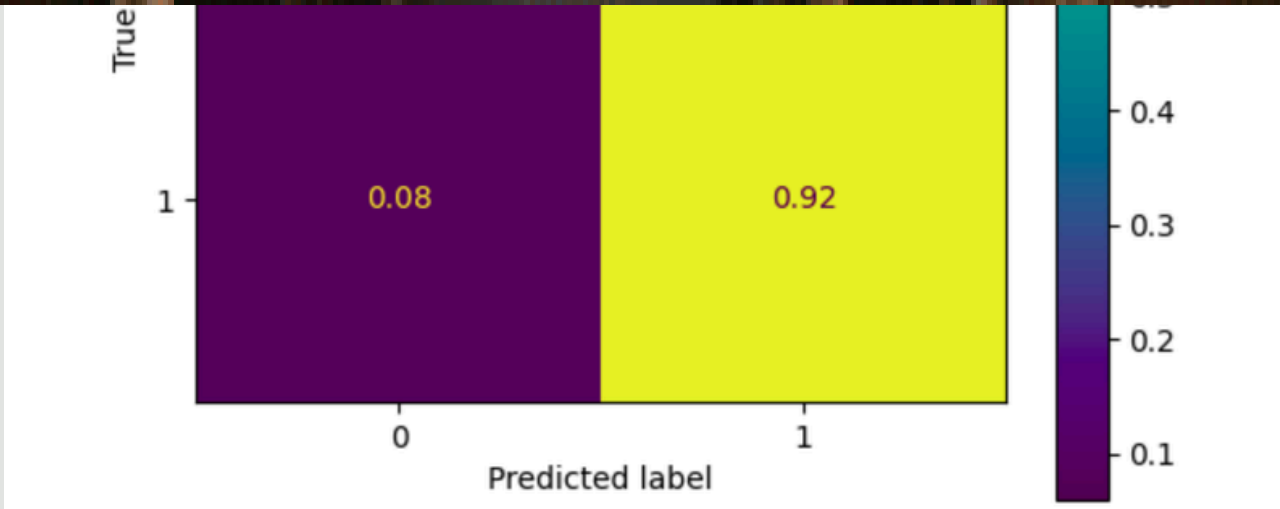




Modelo 0

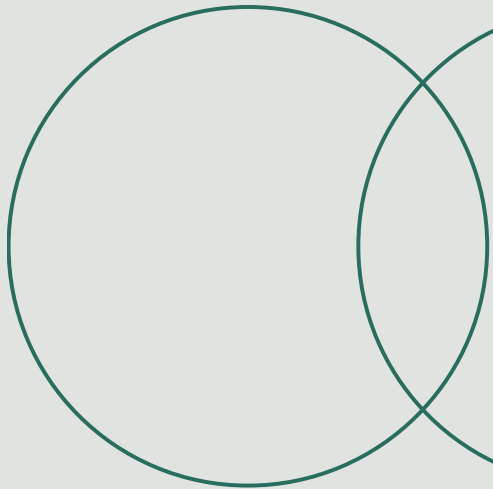


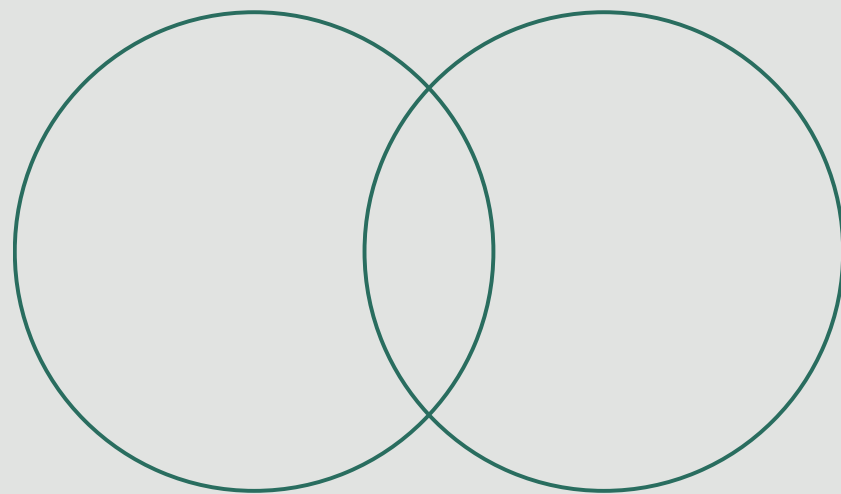
Modelo 2



Modelo 1

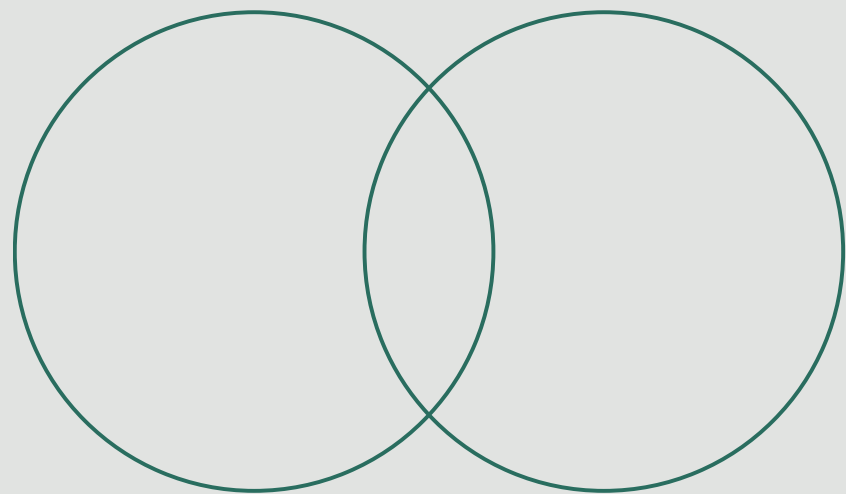
Modelo 3



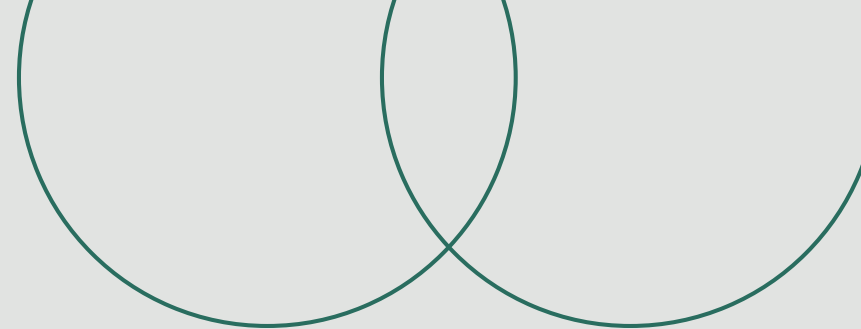


FUTURO

- Intentar sacar estadísticas relevantes a partir de contorno y texturas (media de contornos, radio medio, perímetro, etc)
- Incluir un modelo de validación de entrada (Garantizar que las entradas son de hecho imagenes de histopatología)
- Generar más imagenes para entrenar y mejorar el modelo



MODELOS DESPLIEGABLES (STREAMLIT)



... ¡GRACIAS!

¿Preguntas, comentarios o sugerencias? Estoy a su disposición.

